

Food Ingredients - Eva Hatalčíková, Tomáš Hutňan

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```
[1]: import pandas as pd
import numpy as np
import math
import matplotlib.pyplot as plt
import seaborn as sns

import matplotlib.patches as mpatches
from sklearn.manifold import TSNE
from sklearn.metrics import silhouette_samples, silhouette_score

from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.cluster import KMeans

MANUALLY_PREPROCESSED_CSV_DATA_FILE_PATH: str = "..\\data\\original.csv"

df = pd.read_csv(MANUALLY_PREPROCESSED_CSV_DATA_FILE_PATH, sep=";", decimal=",")
```

1 Exploratory Data Analysis

1.1 Initial Cleanup

Some issues with the data were fixed in excel manually. These include: - Table was not aligned to the top left. - A span of ~400 rows was shifted to the right.

We decided to remove columns with invalid or missing values completely, as there were only 122 such rows. The values appeared to be missing completely at random.

```
[2]: # Coerce invalid values to NaN
object_feature_columns = ["Magnesium_mg", "VitE_mg"]
coerced_count = 0
for col in object_feature_columns:
    df[col] = pd.to_numeric(df[col], errors="coerce")
    coerced_count += np.sum(df[col].isna())
print(f"Coerced {coerced_count} values to NaN in columns_
    ↳{object_feature_columns}.")

# Remove rows with NaN values
row_has_none = df.isna().any(axis=1)
```

```

df = df.dropna()
print(f"Dropped {sum(row_has_none)} rows with missing values.")

# Define numeric columns for future use
num_cols = df.select_dtypes("number").columns

# Coercing duplicate rows
grouped_rows = list(df.groupby("NDB_No", dropna=False, sort=False))
original_row_count = len(df)
merged_rows = []
for ndb_no, group in grouped_rows:
    if len(group) != 1:
        for column in df.columns:
            if group[column].astype(str).nunique() != 1:
                raise ValueError(f"Multiple non-matching rows found for NDB_NO:␣
→{ndb_no}")

        merged_rows.append(group.iloc[0])

df = pd.DataFrame(merged_rows, columns=df.columns).reset_index(drop=True)
print(f"Removed {original_row_count - len(df)} duplicate rows.")

# Convert micro-/mili-grams to grams (automatic, based on suffixes)
suffix_map = {"_mg": 1e-3, "_mcg": 1e-6}
converted = []
for col in list(df.columns):
    for suf, factor in suffix_map.items():
        if col.endswith(suf):
            base = col.rsplit("_", 1)[0]
            new_col = f"{base}_g"
            # ensure numeric before conversion
            df[col] = pd.to_numeric(df[col], errors='coerce')
            df[new_col] = df[col] * factor
            converted.append((col, new_col))
            # drop the original suffixed column after conversion
            df.drop(columns=[col], inplace=True)
            break

print(f"Converted {len(converted)} columns: {converted}")

# Recompute numeric columns
num_cols = df.select_dtypes("number").columns.tolist()

```

Coerced 15 values to NaN in columns ['Magnesium_mg', 'VitE_mg'].

Dropped 122 rows with missing values.

Removed 916 duplicate rows.

Converted 20 columns: [('Calcium_mg', 'Calcium_g'), ('Iron_mg', 'Iron_g'), ('Magnesium_mg', 'Magnesium_g'), ('Phosphorus_mg', 'Phosphorus_g'),

```
(('Potassium_mcg', 'Potassium_g'), ('Sodium_mcg', 'Sodium_g'), ('Zinc_mcg',
'Zinc_g'), ('Copper_mcg', 'Copper_g'), ('Manganese_mcg', 'Manganese_g'),
('Selenium_mcg', 'Selenium_g'), ('VitC_mcg', 'VitC_g'), ('Thiamin_mg',
'Thiamin_g'), ('Riboflavin_mg', 'Riboflavin_g'), ('Niacin_mg', 'Niacin_g'),
('VitB6_mcg', 'VitB6_g'), ('Folate_mcg', 'Folate_g'), ('VitB12_mcg', 'VitB12_g'),
('VitA_mcg', 'VitA_g'), ('VitE_mcg', 'VitE_g'), ('VitD2_mcg', 'VitD2_g'))]
```

1.2 Outliers

Most features have visible outliers, we will need to resolve those later.

```
[3]: # Box plots for all numeric columns
numeric_data = df[num_cols]
n_cols = 3
n_rows = math.ceil(len(num_cols) / n_cols)

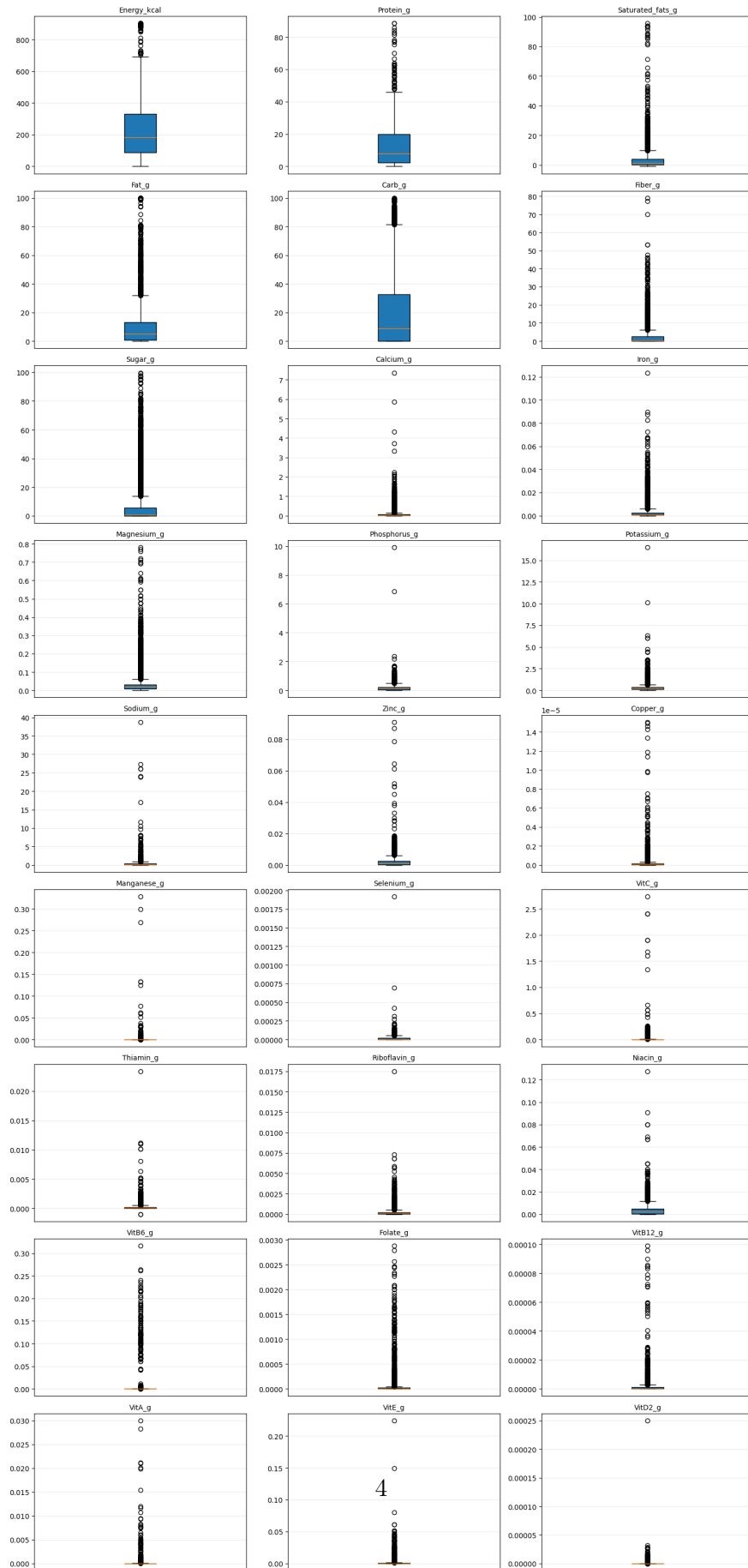
fig, axes = plt.subplots(n_rows, n_cols, figsize=(n_cols * 5, n_rows * 3.5))
axes = axes.flatten() if hasattr(axes, "flatten") else [axes]

for ax, column in zip(axes, num_cols):
    series = numeric_data[column].dropna()
    ax.boxplot(series, vert=True, patch_artist=True)
    ax.set_title(column, fontsize=10)
    ax.set_xticks([])
    ax.grid(axis="y", alpha=0.25)

for ax in axes[len(num_cols):]:
    ax.axis("off")

fig.suptitle("Box plots for all numeric columns", fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```

Box plots for all numeric columns



1.3 Pyramid charts

- Idea taken from Hynek Vaverka group's first presentation.
- We can see most foods respect the minimum calory estimate:

$$\text{kcal} \geq 4 \cdot \text{carbs} + 4 \cdot \text{protein} + 9 \cdot \text{fat}$$

- However, some foods are in forbidden regions of the graphs (a food can't have 100 grams of carbs but 0 calories).

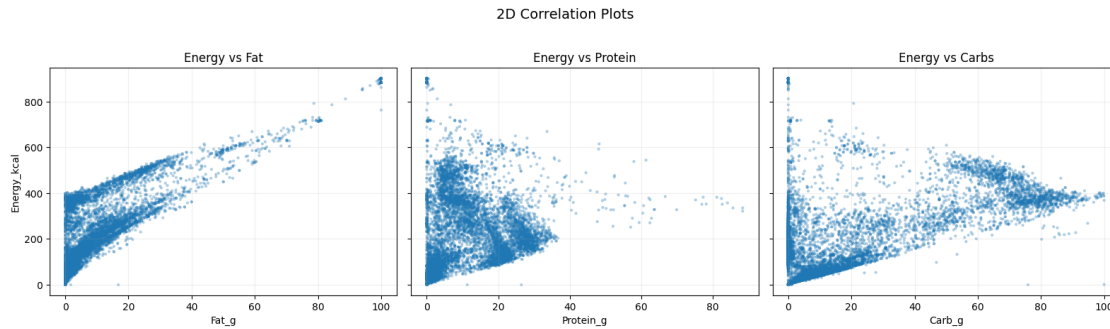
```
[4]: # Simple 2D plots for Energy vs macronutrients (pyramid-like patterns)
pairs = [
    ("Fat_g", "Energy vs Fat"),
    ("Protein_g", "Energy vs Protein"),
    ("Carb_g", "Energy vs Carbs"),
]

fig, axes = plt.subplots(1, 3, figsize=(16, 4.5), sharey=True)

for ax, (x_col, title) in zip(axes, pairs):
    plot_df = df[[x_col, "Energy_kcal"]].copy()
    plot_df[x_col] = pd.to_numeric(plot_df[x_col], errors="coerce")
    plot_df["Energy_kcal"] = pd.to_numeric(plot_df["Energy_kcal"],
    ↪errors="coerce")
    plot_df = plot_df.dropna()

    ax.scatter(
        plot_df[x_col],
        plot_df["Energy_kcal"],
        s=9,
        alpha=0.35,
        edgecolors="none"
    )
    ax.set_xlabel(x_col)
    ax.set_title(title)
    ax.grid(alpha=0.2)

axes[0].set_ylabel("Energy_kcal")
fig.suptitle("2D Correlation Plots", fontsize=14, y=1.03)
plt.tight_layout()
plt.show()
```



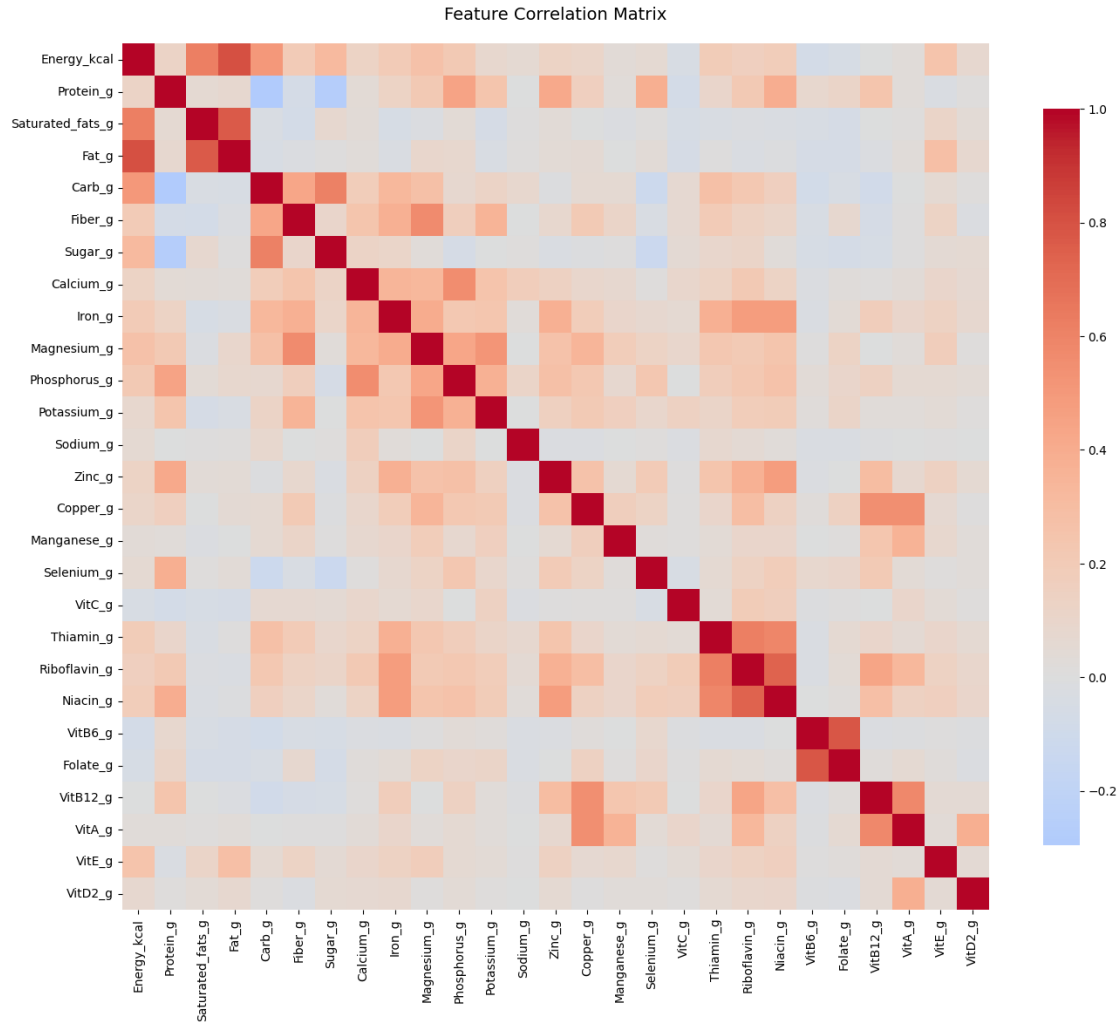
1.4 Feature Correlation

```
[5]: # Compute correlation matrix
corr = df[num_cols].corr()

# Plot heatmap
fig, ax = plt.subplots(figsize=(14, 12))
sns.heatmap(corr, cmap="coolwarm", center=0, square=True, ax=ax,
            cbar_kws={"shrink": 0.8})
ax.set_title("Feature Correlation Matrix", fontsize=14, pad=20)
plt.tight_layout()
plt.show()

# Identify highly correlated pairs (>0.95)
high_corr_pairs = []
for i in range(len(corr.columns)):
    for j in range(i+1, len(corr.columns)):
        if abs(corr.iloc[i, j]) > 0.95:
            high_corr_pairs.append((corr.columns[i], corr.columns[j], corr.
            iloc[i, j]))

print(f"Highly correlated pairs (>0.95): {len(high_corr_pairs)}")
for col1, col2, val in high_corr_pairs[:5]:
    print(f" {col1} <-> {col2}: {val:.3f}")
```



Highly correlated pairs (>0.95): 0

2 Preprocessing

2.1 Outliers

- Clamp negative values
- Use the Atwater nutrient equation to remove extreme outliers

```
[6]: # Start with numeric features, drop ID columns
X = df.select_dtypes(include="number").drop(columns=["NDB_No"], errors="ignore").
    ↪ copy()
print(f"Initial shape: {X.shape}")

# Clamp negative values to 0
neg_count = int((X < 0).sum().sum())
```

```

X = X.clip(lower=0)
print(f"Clamped {neg_count} negative values to 0")

# Remove rows violating the energy equation: kcal >= 4*carbs + 4*protein + 9*fat
ENERGY_THRESHOLD_RATIO = 1.25
ENERGY_THRESHOLD_ABSOLUTE = 50
projected_energy = (4 * X['Carb_g'] + 4 * X['Protein_g'] + 9 * X['Fat_g'])

# Keep only rows where observed energy falls within the expected energy
mask = (X['Energy_kcal'] < projected_energy * ENERGY_THRESHOLD_RATIO) & \
        (X['Energy_kcal'] > projected_energy / ENERGY_THRESHOLD_RATIO) & \
        (abs(X['Energy_kcal'] - projected_energy) > ENERGY_THRESHOLD_ABSOLUTE)
X = X[~mask]

print(f"Removed {int((mask).sum())} rows violating energy equation.")

```

Initial shape: (9213, 27)

Clamped 10 negative values to 0

Removed 57 rows violating energy equation.

2.2 Feature Engineering

- Nutrient ratios
- Mineral and Vitamin densities

```

[7]: # Create engineered features on the cleaned data
X_eng = X.copy()

# If feature is 0, ratio is 0.
X_eng['Protein_per_kcal'] = np.where(X_eng['Energy_kcal'] == 0, 0,
    ↪X_eng['Protein_g'] / X_eng['Energy_kcal'])
X_eng['Fiber_Carb_ratio'] = np.where(X_eng['Carb_g'] == 0, 0, X_eng['Fiber_g'] /
    ↪X_eng['Carb_g'])
X_eng['Saturated_fat_ratio'] = np.where(X_eng['Fat_g'] == 0, 0,
    ↪X_eng['Saturated_fats_g'] / X_eng['Fat_g'])

# Mineral and vitamin densities
minerals = [c for c in X_eng.columns if any(x in c.lower() for x in ['calcium',
    ↪'iron', 'magnesium', 'phosphorus', 'potassium', 'sodium', 'zinc', 'copper',
    ↪'manganese', 'selenium'])]
vitamins = [c for c in X_eng.columns if any(x in c.lower() for x in ['vit',
    ↪'thiamin', 'riboflavin', 'niacin', 'folate'])]

# Scale minerals to 0-1 range first, then sum so trace minerals have equal weight
X_eng['Mineral_density'] = MinMaxScaler().fit_transform(X_eng[minerals]).
    ↪sum(axis=1)

```

```

X_eng['Vitamin_density'] = MinMaxScaler().fit_transform(X_eng[vitamins]).
↳sum(axis=1)

engineered_cols = [c for c in X_eng.columns if c not in X.columns]
print(f"Created {len(engineered_cols)} engineered features: {engineered_cols}")
print(f"Total features: {X_eng.shape[1]}")

```

Created 5 engineered features: ['Protein_per_kcal', 'Fiber_Carb_ratio', 'Saturated_fat_ratio', 'Mineral_density', 'Vitamin_density']
Total features: 32

2.3 Log1p Transform of Higly Skewd Columns

Most foods contain zero or trace amounts of specific micronutrients, while a few “superfoods” possess massive, outlier concentrations. Because K-Means clustering is highly vulnerable to extreme values, we applied this transformation to compress the extreme tails of the data without breaking natural zero-counts. This is a standard practice in nutritional informatics, ensuring that extreme micronutrient outliers (like liver or fortified cereals) do not artificially hijack and collapse the clusters.

```

[8]: # Create log1p-transformed copy (on skewed columns)
X_log1p = X_eng.copy()

# Identify columns with high positive skew
SKEW_THRESHOLD = 1.5
skew_vals = X_eng.skew(numeric_only=True) # Only check non-engineered features
log_cols = skew_vals[skew_vals > SKEW_THRESHOLD].index.tolist()

if log_cols:
    X_log1p[log_cols] = np.log1p(X_log1p[log_cols])
    print(f"Applied log1p to {len(log_cols)} skewed columns: {log_cols}")
else:
    print("No columns above skew threshold")

print(f"Shape: {X_log1p.shape}")

```

Applied log1p to 27 skewed columns: ['Saturated_fats_g', 'Fat_g', 'Fiber_g', 'Sugar_g', 'Calcium_g', 'Iron_g', 'Magnesium_g', 'Phosphorus_g', 'Potassium_g', 'Sodium_g', 'Zinc_g', 'Copper_g', 'Manganese_g', 'Selenium_g', 'VitC_g', 'Thiamin_g', 'Riboflavin_g', 'Niacin_g', 'VitB6_g', 'Folate_g', 'VitB12_g', 'VitA_g', 'VitE_g', 'VitD2_g', 'Fiber_Carb_ratio', 'Mineral_density', 'Vitamin_density']
Shape: (9156, 32)

2.4 Standard Scaler

By standardizing the data, we ensure that every feature is weighted equally. This step is crucial and heavily utilized in dietary epidemiology, as it guarantees that a food’s trace mineral profile has the same mathematical influence on its final classification as its bulk macronutrients.

```
[9]: X_log1p_scaled = X_log1p.copy()

scaler = StandardScaler()
X_log1p_scaled = pd.DataFrame(
    scaler.fit_transform(X_log1p),
    index=X_log1p.index,
    columns=X_log1p.columns,
)

X_preprocessed = X_log1p_scaled.copy()

print(f"Shape after preprocessing: {X_preprocessed.shape}")
```

Shape after preprocessing: (9156, 32)

3 Models

3.1 Shared Functions

```
[10]: def evaluate_and_visualize_clusters(raw_data, scaled_data, labels):
    """
    Takes readable data (for stats) and scaled data (for distance metrics),
    plus cluster labels. Plots a 3x2 grid:
    Top: Pie Chart | t-SNE
    Mid: Silhouette Plot | Macronutrient Pyramid
    Bot: Radar Chart | Violin Plot
    """

    # 1. Prepare the DataFrames
    df_raw = pd.DataFrame(raw_data).copy()
    df_scaled = pd.DataFrame(scaled_data).copy()

    df_raw['Cluster'] = labels
    unique_labels = np.unique(labels)
    n_clusters = len(unique_labels)

    # =====
    # GLOBAL COLOR & ALIAS MAPPING
    # =====
    aliases = {label: f"C{i}" for i, label in enumerate(unique_labels)}
    df_raw['Alias'] = df_raw['Cluster'].map(aliases)

    colors = sns.color_palette("husl", n_clusters)
    palette = dict(zip(unique_labels, colors))
    palette_alias = {aliases[k]: v for k, v in palette.items()}

    # =====
    # 2. CALCULATE AND PRINT STATISTICS
```

```

# =====
print("=" * 50)
print("CLUSTER SIZE STATISTICS")
print("=" * 50)

cluster_counts = df_raw['Cluster'].value_counts().sort_index()
cluster_percentages = (cluster_counts / len(df_raw)) * 100

stats_df = pd.DataFrame({
    'Alias': [aliases[idx] for idx in cluster_counts.index],
    'Size (Rows)': cluster_counts,
    'Percentage (%)': cluster_percentages.round(2)
})
print(stats_df.to_string())
print("\n")

# =====
# 3. DIMENSIONALITY REDUCTION & SILHOUETTE
# =====
tsne = TSNE(n_components=2, random_state=42, n_jobs=-1)
tsne_result = tsne.fit_transform(df_scaled)

# Calculate global and per-sample silhouette scores
silhouette_avg = silhouette_score(df_scaled, labels)
sample_silhouette_values = silhouette_samples(df_scaled, labels)

# =====
# 4. BUILD THE 3x2 DASHBOARD
# =====
fig = plt.figure(figsize=(18, 22))
fig.suptitle('Comprehensive Cluster Analysis Dashboard', fontsize=24,
fontweight='bold', y=0.98)

# --- Plot 1 (Top Left): Pie Chart ---
ax1 = fig.add_subplot(3, 2, 1)
pie_colors = [palette[val] for val in cluster_counts.index]
ax1.pie(
    cluster_counts, labels=[aliases[idx] for idx in cluster_counts.index],
    autopct='%1.1f%%', colors=pie_colors, startangle=140,
    wedgeprops={'edgecolor': 'white', 'linewidth': 1.5}
)
ax1.set_title('Cluster Distribution', fontsize=16)

# --- Plot 2 (Top Right): t-SNE ---
ax2 = fig.add_subplot(3, 2, 2)
sns.scatterplot(
    x=tsne_result[:, 0], y=tsne_result[:, 1],

```

```

        hue=df_raw['Alias'], palette=palette_alias, alpha=0.6, s=40, ax=ax2,
→legend=False
    )
    ax2.set_title('Local Neighborhoods (t-SNE)', fontsize=16)
    ax2.set_xlabel('t-SNE Dimension 1')
    ax2.set_ylabel('t-SNE Dimension 2')

    # --- Plot 3 (Mid Left): Silhouette Plot ---
    ax3 = fig.add_subplot(3, 2, 3)
    y_lower = 10

    for label in unique_labels:
        # Aggregate and sort silhouette scores for this cluster
        ith_cluster_silhouette_values = sample_silhouette_values[labels == label]
        ith_cluster_silhouette_values.sort()

        size_cluster_i = ith_cluster_silhouette_values.shape[0]
        y_upper = y_lower + size_cluster_i

        color = palette[label]
        ax3.fill_betweenx(
            np.arange(y_lower, y_upper), 0, ith_cluster_silhouette_values,
            facecolor=color, edgecolor=color, alpha=0.7
        )

        # Label the clusters in the middle of their silhouette "blade"
        ax3.text(-0.05, y_lower + 0.5 * size_cluster_i, aliases[label],
→fontsize=12, va='center', ha='right')
        y_lower = y_upper + 10 # Gap between clusters

    ax3.set_title("Silhouette Plot of Clusters", fontsize=16)
    ax3.set_xlabel("Silhouette Coefficient Values")
    ax3.set_ylabel("Cluster Alias (Thickness = Size)")
    ax3.axvline(x=silhouette_avg, color="red", linestyle="--") # Global average
→line
    ax3.set_yticks([]) # Clear the y-axis ticks to keep it clean

    # --- Plot 4 (Mid Right): Macronutrient Caloric Breakdown ---
    ax4 = fig.add_subplot(3, 2, 4)
    medians = df_raw.groupby('Cluster').median(numeric_only=True)

    macro_cals = pd.DataFrame({
        'Carb_cals': medians['Carb_g'] * 4,
        'Protein_cals': medians['Protein_g'] * 4,
        'Fat_cals': medians['Fat_g'] * 9
    })
    macro_totals = macro_cals.sum(axis=1).replace(0, 1)

```

```

macro_pct = macro_cals.div(macro_totals, axis=0) * 100
macro_pct.index = [aliases[label] for label in macro_pct.index]

macro_pct.plot(
    kind='bar', stacked=True, ax=ax4,
    color=['#FFB347', '#FF6961', '#77DD77'], edgecolor='black', linewidth=1.2
)
ax4.set_title('Median Macronutrient Energy Breakdown', fontsize=16)
ax4.set_ylabel('% of Calculated Calories')
ax4.legend(['Carbs', 'Protein', 'Fat'], loc='lower center',
    ↳bbox_to_anchor=(0.5, 1.02), ncol=3)
ax4.tick_params(axis='x', rotation=0)

# --- Plot 5 (Bottom Left): Radar Chart (Spider Plot) ---
ax5 = fig.add_subplot(3, 2, 5, polar=True)
radar_features = ['Carb_g', 'Protein_g', 'Fat_g', 'Fiber_g', 'Sugar_g']
radar_data = medians[radar_features]

scaler = MinMaxScaler()
radar_scaled = pd.DataFrame(scaler.fit_transform(radar_data),
    ↳columns=radar_features, index=radar_data.index)

N = len(radar_features)
angles = [n / float(N) * 2 * np.pi for n in range(N)]
angles += angles[:1]
ax5.set_theta_offset(np.pi / 2)
ax5.set_theta_direction(-1)
ax5.set_xticks(angles[:-1])
ax5.set_xticklabels(radar_features)

for cluster_label in radar_scaled.index:
    values = radar_scaled.loc[cluster_label].values.flatten().tolist()
    values += values[:1]
    ax5.plot(angles, values, linewidth=2, linestyle='solid',
    ↳color=palette[cluster_label])
    ax5.fill(angles, values, color=palette[cluster_label], alpha=0.15)

ax5.set_title('Normalized Macro Shape (Radar)', fontsize=16, pad=20)

# --- Plot 6 (Bottom Right): Violin Plot ---
ax6 = fig.add_subplot(3, 2, 6)
sns.violinplot(
    x='Alias', y='Energy_kcal', data=df_raw,
    hue='Alias', palette=palette_alias, density_norm='width',
    ↳inner='quartile', ax=ax6, legend=False
)
ax6.set_title('Energy (kcal) Distribution Spread', fontsize=16)

```

```

ax6.set_ylabel('Energy (kcal / 100g)')
ax6.set_xlabel('Cluster Alias')

# =====
# GLOBAL MASTER LEGEND
# =====
global_patches = [mpatches.Patch(color=palette[label],
→label=f"{aliases[label]}: {label}") for label in unique_labels]
legend_cols = min(4, len(unique_labels))
fig.legend(handles=global_patches, loc='lower center', bbox_to_anchor=(0.5,
→0.01), ncol=legend_cols, fontsize=13)

plt.tight_layout(rect=[0, 0.05, 1, 0.95])
plt.show()

```

3.2 Baseline (Random)

```

[11]: k = 5
n_samples = len(X_preprocessed)

np.random.seed(42) # Set seed for reproducibility
random_labels = np.random.randint(0, k, size=n_samples)

baseline_silhouette = silhouette_score(X_preprocessed, random_labels)

print(f"Baseline Silhouette Score : {baseline_silhouette:.4f}")
evaluate_and_visualize_clusters(X_eng, X_preprocessed, random_labels)

```

Baseline Silhouette Score : -0.0187

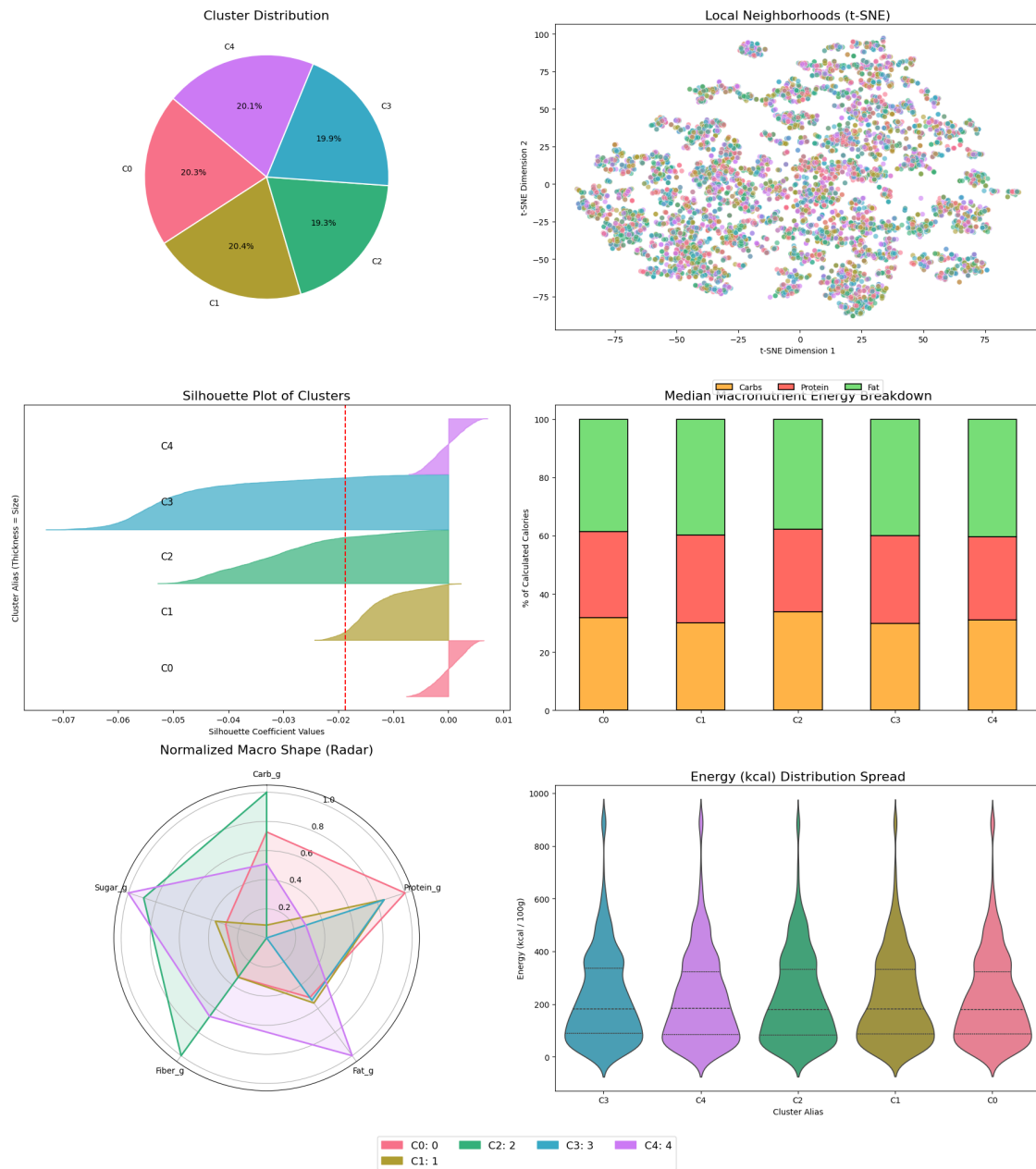
=====

CLUSTER SIZE STATISTICS

=====

	Alias	Size (Rows)	Percentage (%)
Cluster			
0	C0	1855	20.26
1	C1	1867	20.39
2	C2	1770	19.33
3	C3	1823	19.91
4	C4	1841	20.11

Comprehensive Cluster Analysis Dashboard



3.3 DBSCAN (Euka)

The main goal of this section of the project was to identify an appropriate clustering structure using DBSCAN for the nutritional dataset. The focus was on tuning the key DBSCAN parameters — particularly `eps` and `min_samples` — to discover meaningful groupings of foods based on their nutrient profiles while minimizing noise and avoiding over- or under-clustering.

According to the research by Yongyu Wang, DBSCAN faces significant challenges in high-dimensional spaces because its computational complexity scales with the number of dimensions (D), making distance calculations increasingly resource-intensive and often computationally infeasible for large-scale analysis. While preprocessing is necessary to address these efficiency bottlenecks, Wang emphasizes that it also serves a critical qualitative purpose: by using methods like spectral data compression to remove substantial redundancy and noise, the algorithm can focus on the most essential structural information. As a result, this type of processing does not merely accelerate the execution but significantly enhances the quality and accuracy of the final clustering solutions.

According to documentation, PCA algorithm can reduce dimensionality, but reducing it too far (e.g., to 10 or fewer dimensions) often discards the very variance needed to distinguish clusters - due to its linear nature. UMAP is designed for non-linear manifold-aware dimension reduction, which can compress data into a very small number of dimensions (like 10) while preserving the cluster structure. Therefore, in this section UMAP preprocessing will be applied.

3.3.1 Reducing dimensions

```
[12]: from umap import UMAP

# We use parameters optimized for clustering (pre-clustering)
umap_precluster = UMAP(
    n_neighbors=100,      # High value (30-100) to focus on global structure
    min_dist=0.0,        # Set to 0 to pack points together densely for
    # clustering
    n_components=10,      # Reduce to 10 dimensions to retain more variance than
    # 2D
    metric="euclidean",   # Standard distance metric used in the source examples
    random_state=43       # Ensures reproducibility of the results
)

# Transform the data to the new 10-dimensional space
X_umap = umap_precluster.fit_transform(X_preprocessed)

print(f"Original shape: {X_preprocessed.shape}")
print(f"Reduced shape: {X_umap.shape}")
```

```
c:\Users\tomas\Projects\courses\ib031\ib031-food-ingredients\.venv\Lib\site-
packages\umap\umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by
setting random_state. Use no seed for parallelism.
```

```
warn(
```

```
Original shape: (9156, 32)
```

```
Reduced shape: (9156, 10)
```

3.3.2 Visualization

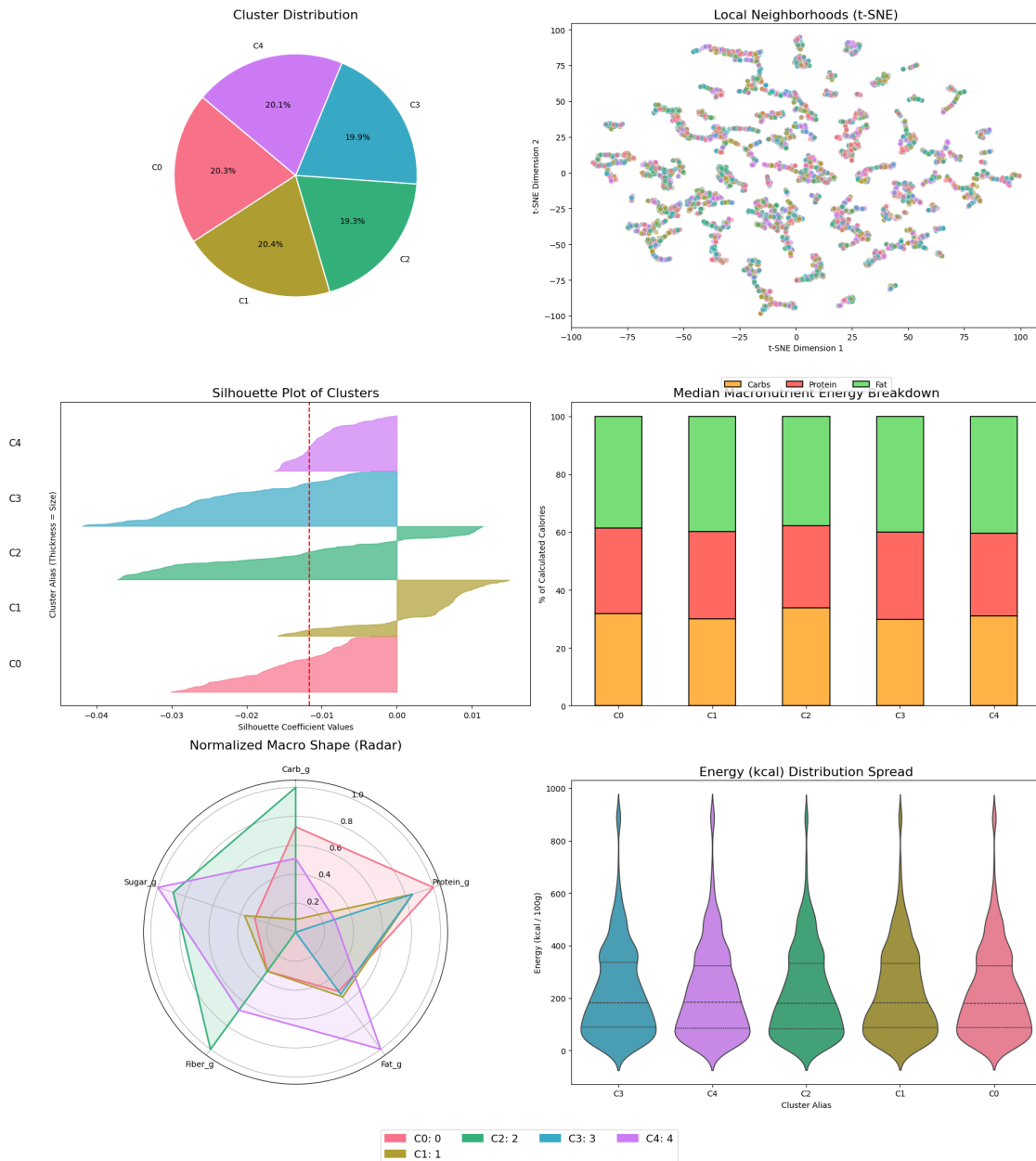
In this section, the dimensionally reduced dataset will be visualized using t-SNE and UMAP to examine potential clustering structures and identify natural groupings within the data.

```
[13]: evaluate_and_visualize_clusters(X_eng, X_umap, random_labels)
```

```
=====
CLUSTER SIZE STATISTICS
=====
```

	Alias	Size (Rows)	Percentage (%)
Cluster			
0	C0	1855	20.26
1	C1	1867	20.39
2	C2	1770	19.33
3	C3	1823	19.91
4	C4	1841	20.11

Comprehensive Cluster Analysis Dashboard



After visualizing the dataset using t-SNE, the resulting embeddings revealed several visually distinct and well-structured groupings within the nutritional data. These projections suggest that meaningful underlying patterns exist among foods based on their nutrient compositions, indicating strong potential for successful clustering using density-based methods such as DBSCAN or HDBSCAN. The visual separation observed in the lower-dimensional representations provides a promising foundation for identifying natural food clusters and nutritional profiles.

3.3.3 Training DBSCAN

```
[14]: import numpy as np
import pandas as pd
from sklearn.cluster import DBSCAN
from sklearn.metrics import silhouette_score
from umap import UMAP

RANDOM_STATE = 43
SEARCH_SIZE = X_umap.shape[0] # Use all samples for search if dataset is small
VIS_SIZE = X_umap.shape[0] # Use all samples for visualization if dataset is
    → small

search_size = min(SEARCH_SIZE, len(X_umap))
rng_search = np.random.RandomState(RANDOM_STATE)
search_positions = rng_search.choice(len(X_umap), size=search_size,
    → replace=False)

search_X_umap = X_umap[search_positions] # random samples

search_X_eval = X_preprocessed.values[search_positions]

# DBSCAN GRID SEARCH
param_grid = {
    "eps": [1.0, 1.2, 1.5, 2.0],
    "min_samples": [15, 20, 22, 25]
}

results = []
best_score = -1
best_params = None

print("\nRunning DBSCAN parameter sweep...")
for eps in param_grid["eps"]:
    for min_samples in param_grid["min_samples"]:
        model = DBSCAN(eps=eps, min_samples=min_samples)
        labels = model.fit_predict(search_X_umap)

        n_clusters = len(set(labels)) - (1 if -1 in labels else 0)

        score = 0
        if n_clusters > 1:
            # Evaluating on original 30D space for accuracy

            non_noise = labels != -1
```

```

        filtered_labels = labels[non_noise]

        if len(set(filtered_labels)) > 1:

            score = silhouette_score(
                search_X_umap[non_noise],
                filtered_labels
            )
            results.append({"eps": eps, "min_samples": min_samples, "n_clusters":
↪ n_clusters, "score": score})

            if score > best_score:
                best_score = score
                best_params = {"eps": eps, "min_samples": min_samples}
                best_labels = labels

results_df = pd.DataFrame(results).sort_values("score", ascending=False)
print("\nTOP CONFIGURATIONS:")
print(results_df.head(5).to_string(index=False))

print(f"\nBest Configuration: {best_params}")

```

Running DBSCAN parameter sweep...

TOP CONFIGURATIONS:

eps	min_samples	n_clusters	score
2.0	25	4	0.472136
2.0	22	4	0.472136
2.0	20	4	0.472136
2.0	15	4	0.472136
1.5	25	4	0.472136

Best Configuration: {'eps': 1.5, 'min_samples': 15}

[15]: evaluate_and_visualize_clusters(X_eng, search_X_umap, best_labels)

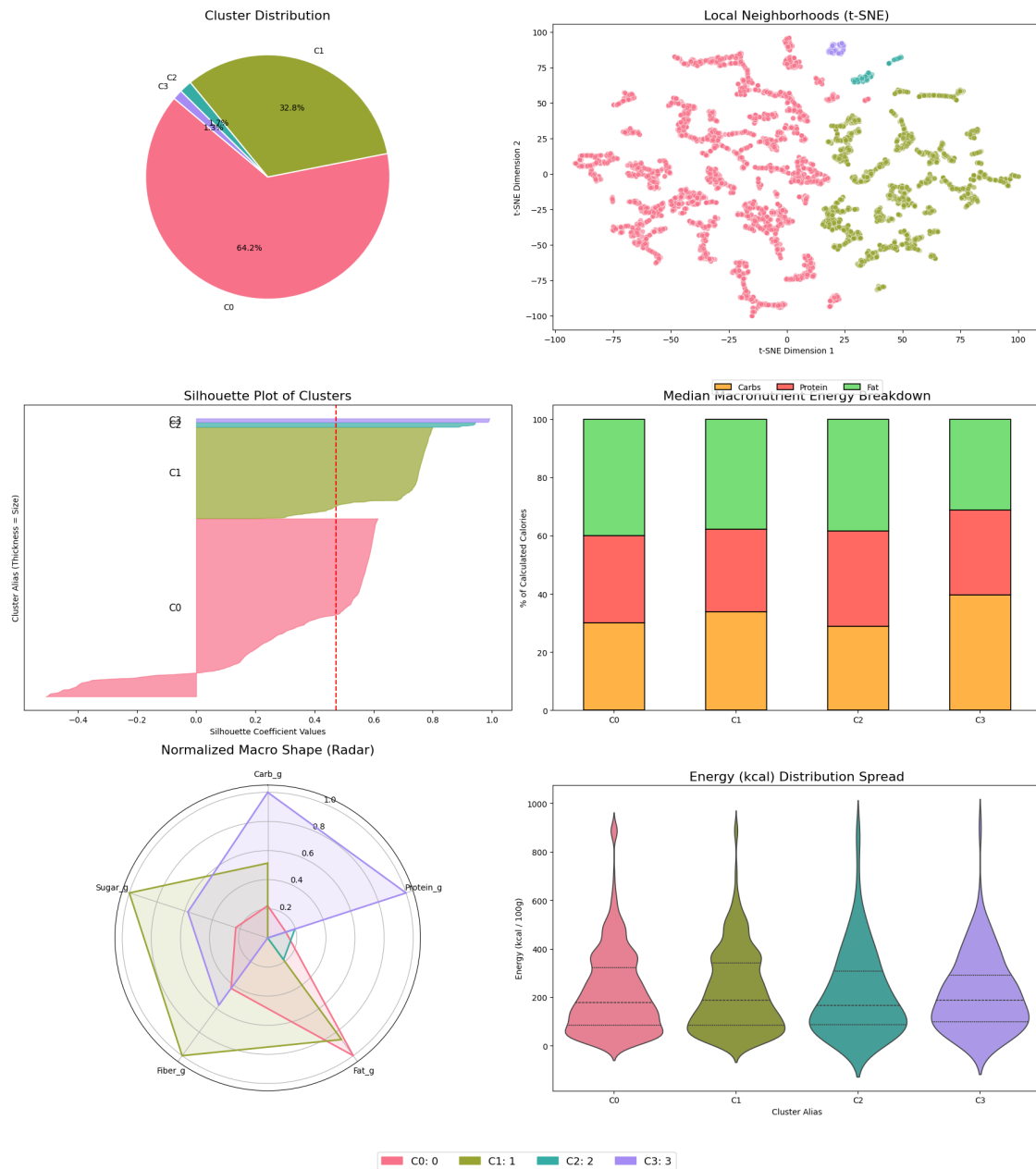
```

=====
CLUSTER SIZE STATISTICS
=====

```

	Alias	Size (Rows)	Percentage (%)
Cluster			
0	C0	5878	64.20
1	C1	3007	32.84
2	C2	152	1.66
3	C3	119	1.30

Comprehensive Cluster Analysis Dashboard



Despite exploring various parameter combinations through techniques like grid search, the DBSCAN algorithm frequently produces unsatisfactory clusters. Even though the best result was scored at pleasing 0.472136, the visualized result is underwhelming and not accurate, as 4 clusters is not sufficient.

Higher score than 0.472136 was achieved by increasing the eps metric, but the result was vague - with only few large clusters.

In this section, the algorithm HDBSCAN will be utilised on the same dataset, to study its results and later compare them to DBSCAN.

```
[16]: from hdbscan import HDBSCAN

# 1. Define the grid from your source [9]
param_grid = {
    "min_cluster_size": [10,12,15,20 ],
    "min_samples": [ 25,30, 35],
    "cluster_selection_epsilon": [0.1,0.15,0.2,0.5],
}

search_size = min(SEARCH_SIZE, len(X_umap))
rng_search = np.random.RandomState(RANDOM_STATE)
search_positions = rng_search.choice(len(X_umap), size=search_size,
    ↪replace=False)

search_X_umap = X_umap[search_positions] # random samples

search_X_eval = X_log1p_scaled.values[search_positions]
results = []
best_score = -1
best_params = None

# 2. Run the sweep on your 10D UMAP search sample [10]
# Assuming 'search_X' is your UMAP-reduced data
for mcs in param_grid["min_cluster_size"]:
    for ms in param_grid["min_samples"]:
        for cse in param_grid["cluster_selection_epsilon"]:
            model = HDBSCAN(
                min_cluster_size=mcs,
                min_samples=ms,
                cluster_selection_epsilon=cse,
                cluster_selection_method="eom"
            )
            labels = model.fit_predict(search_X_umap)
            n_clusters = len(set(labels)) - (1 if -1 in labels else 0)

            score = 0
            if n_clusters > 1:
                # Evaluating on original 30D space for accuracy

                if len(set(filtered_labels)) > 1:

                    score = silhouette_score(
                        search_X_umap,
```

```

        labels
    )
    results.append({"min_cluster_size": mcs, "min_samples": ms,
↪ "cluster_selection_epsilon": cse, "n_clusters": n_clusters, "score": score})

    if score > best_score:
        best_score = score
        best_params = {"min_cluster_size": mcs, "min_samples": ms,
↪ "cluster_selection_epsilon": cse}
        best_labels = labels

results_df = pd.DataFrame(results).sort_values("score", ascending=False)
print("\nTOP CONFIGURATIONS:")
print(results_df.head(5).to_string(index=False))

print(f"\nBest Configuration: {best_params}")

```

TOP CONFIGURATIONS:

min_cluster_size	min_samples	cluster_selection_epsilon	n_clusters	score
20	30	0.20	37	0.343767
20	30	0.10	39	0.337640
20	30	0.15	39	0.337640
15	30	0.20	42	0.305524
15	30	0.15	45	0.296715

Best Configuration: {'min_cluster_size': 20, 'min_samples': 30,
'cluster_selection_epsilon': 0.2}

[17]: evaluate_and_visualize_clusters(X_eng, search_X_umap, best_labels)

```

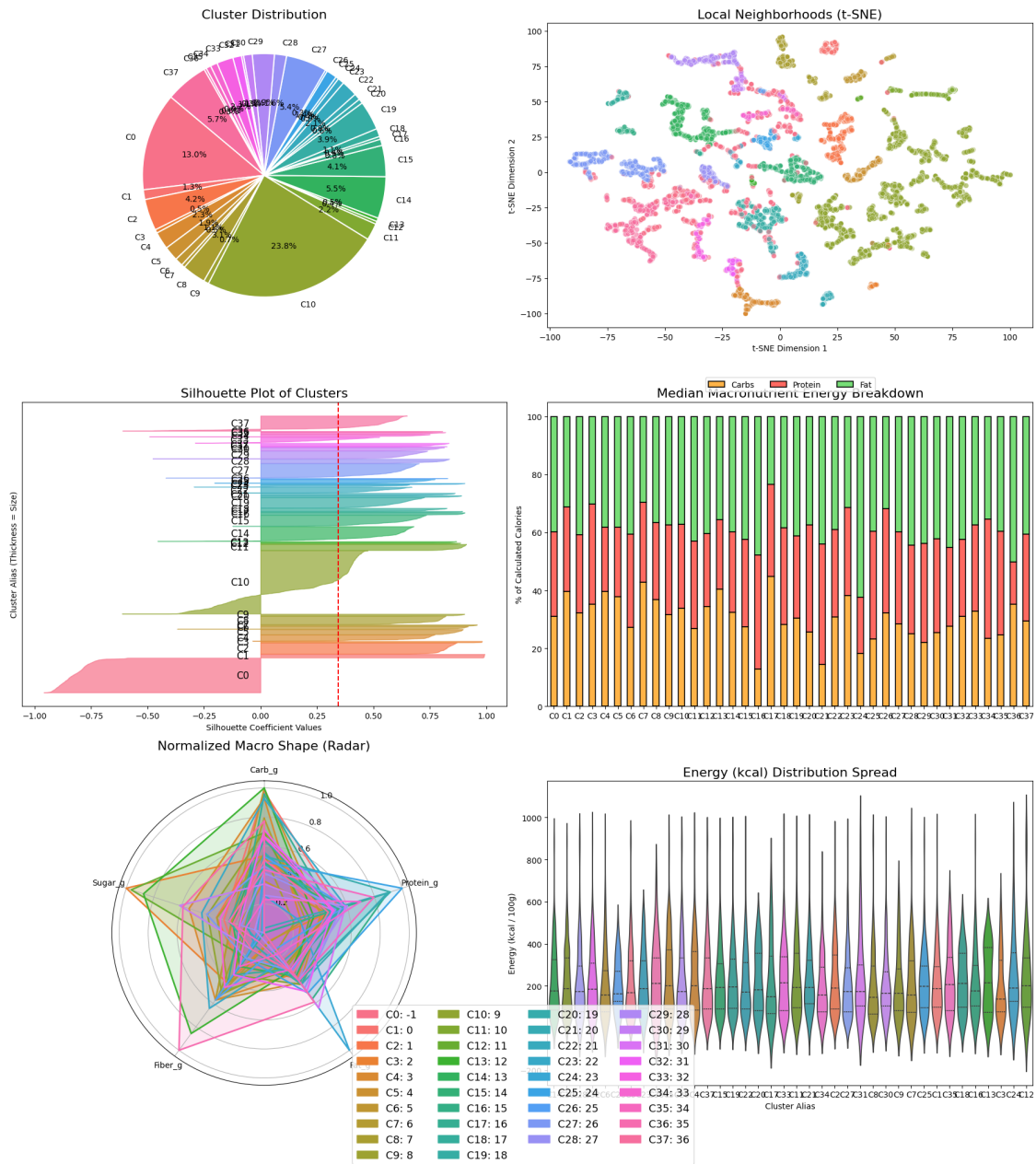
=====
CLUSTER SIZE STATISTICS
=====

```

Cluster	Alias	Size (Rows)	Percentage (%)
-1	C0	1192	13.02
0	C1	119	1.30
1	C2	383	4.18
2	C3	43	0.47
3	C4	212	2.32
4	C5	177	1.93
5	C6	98	1.07
6	C7	42	0.46
7	C8	286	3.12
8	C9	62	0.68
9	C10	2176	23.77
10	C11	205	2.24

11	C12	34	0.37
12	C13	42	0.46
13	C14	508	5.55
14	C15	379	4.14
15	C16	77	0.84
16	C17	34	0.37
17	C18	99	1.08
18	C19	355	3.88
19	C20	56	0.61
20	C21	77	0.84
21	C22	196	2.14
22	C23	78	0.85
23	C24	35	0.38
24	C25	127	1.39
25	C26	25	0.27
26	C27	497	5.43
27	C28	147	1.61
28	C29	268	2.93
29	C30	105	1.15
30	C31	25	0.27
31	C32	105	1.15
32	C33	200	2.18
33	C34	85	0.93
34	C35	59	0.64
35	C36	28	0.31
36	C37	520	5.68

Comprehensive Cluster Analysis Dashboard



As we can see, the HDBSCAN yielded very different results as DBSCAN.

Finding the right combination of parameters was challenging, since increasing the cluster_selection_epsilon or min_cluster_size can artificially inflate silhouette scores by forcing the algorithm to merge smaller density peaks into a few massive groupings. While this “optimization” might suggest a better result, it often leads to fewer clusters that are “visibly incorrect”.

Similarly, adjusting min_samples to higher values can make the model overly conservative, pushing

significant portions of the data into the noise category to satisfy the algorithm’s goal of “not being wrong”. This results in an “underwhelming” clustering that ignores the natural gaps and “manifold boundaries” visible in t-SNE embeddings, effectively trading the recovery of true data structure for a higher, but ultimately misleading, mathematical validation score.

3.3.4 Comparison for HDBSCAN

```
[18]: from hdbscan import HDBSCAN
      from sklearn.manifold import TSNE
      from sklearn.metrics import silhouette_score

      import matplotlib.pyplot as plt
      import numpy as np

      configs = [
          # BASELINE
          {
              "title": "Baseline",
              "min_cluster_size": 5,
              "min_samples": 12,
              "eps": 0.5
          },

          # EPSILON VARIATIONS
          {
              "title": "Lower eps",
              "min_cluster_size": 5,
              "min_samples": 12,
              "eps": 0.2
          },

          {
              "title": "Higher eps",
              "min_cluster_size": 5,
              "min_samples": 12,
              "eps": 1.0
          },

          # MIN CLUSTER SIZE VARIATIONS
          {
              "title": "Lower min_cluster_size",
              "min_cluster_size": 3,
              "min_samples": 12,
              "eps": 0.5
          },

          {
```

```

        "title": "Higher min_cluster_size",
        "min_cluster_size": 15,
        "min_samples": 12,
        "eps": 0.5
    },

    # MIN SAMPLES VARIATIONS
    {
        "title": "Lower min_samples",
        "min_cluster_size": 5,
        "min_samples": 5,
        "eps": 0.5
    },

    {
        "title": "Higher min_samples",
        "min_cluster_size": 5,
        "min_samples": 20,
        "eps": 0.5
    },

    # COMBINED VARIATIONS
    {
        "title": "Loose clustering",
        "min_cluster_size": 3,
        "min_samples": 5,
        "eps": 1.0
    },

    {
        "title": "Strict clustering",
        "min_cluster_size": 15,
        "min_samples": 20,
        "eps": 0.2
    }
]

print("Generating t-SNE embedding...")

tsne = TSNE(
    n_components=2,
    perplexity=30,
    init="pca",
    learning_rate="auto",
    random_state=42
)

```

```

embedding = tsne.fit_transform(X_umap)

fig, axes = plt.subplots(
    3,
    3,
    figsize=(18, 18)
)

axes = axes.flatten()

for idx, config in enumerate(configs):

    model = HDBSCAN(
        min_cluster_size=config["min_cluster_size"],
        min_samples=config["min_samples"],
        cluster_selection_epsilon=config["eps"],
        cluster_selection_method="eom"
    )
    labels = model.fit_predict(X_umap)

    n_clusters = len(set(labels)) - (1 if -1 in labels else 0)
    noise_pct = np.mean(labels == -1) * 100

    sil_score = np.nan
    if n_clusters > 1:

        non_noise = labels != -1
        filtered_labels = labels[non_noise]

        if len(set(filtered_labels)) > 1:

            sil_score = silhouette_score(
                X_umap[non_noise],
                filtered_labels
            )

    display_labels = labels.copy().astype(int)
    if (display_labels == -1).any():

        display_labels[
            display_labels == -1
        ] = display_labels.max() + 1

```

```

ax = axes[idx]
ax.scatter(
    embedding[:, 0],
    embedding[:, 1],
    c=display_labels,
    cmap="tab20",
    s=10,
    alpha=0.8
)

sil_text = (
    f"{sil_score:.3f}"
    if not np.isnan(sil_score)
    else "NaN"
)

ax.set_title(
    (
        f"{config['title']}\n"
        f"min_cluster_size={config['min_cluster_size']}, "
        f"min_samples={config['min_samples']}, "
        f"eps={config['eps']}\n"
        f"clusters={n_clusters}, "
        f"noise={noise_pct:.1f}%\n"
        f"silhouette={sil_text}"
    ),
    fontsize=10,
    fontweight="bold"
)

ax.set_xticks([])
ax.set_yticks([])
ax.grid(alpha=0.2)

plt.suptitle(
    "HDBSCAN Parameter Sensitivity Comparison (t-SNE Visualization)",
    fontsize=18,
    fontweight="bold"
)

plt.tight_layout()
plt.show()

```

Generating t-SNE embedding...



We observe that even small changes in hyperparameters can lead to substantially different clustering results. This makes it difficult to identify a stable and clearly optimal configuration. While the silhouette score tends to increase with a larger number of clusters, solutions with 50+ clusters are not practically useful for interpretation, as they over-segment the data into overly fine-grained groups.

On the other hand, when the number of clusters is constrained to around 10, the t-SNE visualizations show that a large proportion of the data collapses into one dominant cluster, with only a few smaller clusters separating out. This indicates under-segmentation, where meaningful structure is not fully captured.

Overall, there is a clear trade-off between cluster granularity and interpretability, with no single configuration providing both strong separation and meaningful, human-interpretable groupings.

Lets evaluate the best configuration of HDBSCAN:

```
[19]: # Reattach original dataframe
df_results = X_eng.copy()

best_model = model = HDBSCAN(
    min_cluster_size=best_params["min_cluster_size"],
    min_samples=best_params["min_samples"],
    cluster_selection_epsilon=best_params["cluster_selection_epsilon"],
    cluster_selection_method="eom"
)

df_results['Cluster'] = best_labels
df_results['Descrip'] = df.loc[df_results.index, 'Descrip']
df_results['NDB_No'] = df.loc[df_results.index, 'NDB_No']

# Reorder the columns
cols = ['NDB_No', 'Descrip', 'Cluster'] + [c for c in df_results.columns if c
    ↪not in ['NDB_No', 'Descrip', 'Cluster']]
df_results = df_results[cols]

# Print 10 random foods from each cluster:
for i in range(df_results['Cluster'].nunique()):
    print("\n=====")
    print(f"CLUSTER {i} SAMPLES")
    print("=====")

    # Min in case a cluster is tiny
    sample_size = min(10, len(df_results[df_results['Cluster'] == i]))
    samples = df_results[df_results['Cluster'] == i]['Descrip'].
    ↪sample(sample_size, random_state=42)

    for food in samples.values:
        print(f"- {food}")
```

```
=====
CLUSTER 0 SAMPLES
=====
- pie choc mousse prep from mix no-bake type
- lamb aus imp frsh shldr blade lfat 18"fat raw
- ice creams french vanilla soft-serve
- alcoholic bev beer lt lo carb
- beverages energy drk rockstar sugar free
- fast foods griddle cake sndwch egg chs sausage
- collards frz chopd ckd bld drnd wosalt
- beverages choc pdr no sugar added
- chicken roasting meat only ckd rstcd
- chicken broilers or fryers back meat only raw
```

=====
CLUSTER 1 SAMPLES
=====

- beef rnd eye of rnd steak bnless ln ft 0" fat all grds raw
- rice white medium-grain enr ckd
- milk dry nonfat inst wo added vit a vitamin d
- doughnuts cake-type pln sugared or glazed
- beef rib whl (ribs 6-12) lnfat 18"fat all grds raw
- campbell's healthy request homestyle chick noodle soup cond
- peanuts all types ckd bld wsalt
- babyfood dssrt cherry vanilla pudd str
- sunflower sd flr part defatted
- fast foods submarine sndwch bacon lettuce tomato wht brd

=====
CLUSTER 2 SAMPLES
=====

- babyfood dinner chick stew todd
- drumstick pods raw
- beverages tea inst decaffeinated lemon diet
- beef australian imp grass-fed external fat raw
- whey sweet dried
- pumpkin canned with salt
- turkey breast from whl bird meat only w added soln raw
- cereals rte kellogg kellogg's raisin bran w cranberries
- sweet potato leaves ckd stmd w salt
- inf formula pbm products store brand soy pdr

=====
CLUSTER 3 SAMPLES
=====

- potatoes mshd dehyd prep frm flks wo milk whl milk marg add
- yellowtail mxd sp raw
- beverages fruit punch juc drk frz conc
- chicken heart all classes ckd simmr
- snacks candy rolls yogurt-covered fruit flav w hi vit c
- grapes red or grn (euro type such as thompson seedless) raw
- pork cured ham h2o product slice bnless ln htd pan-broil
- falafel home-prepared
- okra frz ckd bld drnd w salt
- supper bakes meal chick w stuffing (chick not included)

=====
CLUSTER 4 SAMPLES
=====

- plums dried (prunes) unckd
- cheese cottage wveg

- soup shark fin rest-prep
- fat duck
- capsicum yellow
- chicken broilers or fryers wing meatskn ckd fried flr
- veal breast plate half bnless lnfat ckd brsd
- potatoes hash brown frz pln unprep
- jambu fruit ripe
- gravy au jus dry

=====

CLUSTER 5 SAMPLES

=====

- puddings rice dry mix prep w whl milk
- rhubarb frozen uncooked
- duck yng duckl dom wh pekin brst meat bnless ckd woskn brld
- perinkilichai
- beef rnd top rnd steak ln 18" fat all grds ckd brld
- alcoholic bev wine table red zinfandel
- beverages carb club soda
- keebler cinn toast grahams
- soya bean white
- chrysanthemum garland ckd bld drnd wosalt

=====

CLUSTER 6 SAMPLES

=====

- veal australian shank fore bone-in ln raw
- salad drsng french drsng comm reg
- game meat muskrat ckd rstcd
- sesame flour low-fat
- lamb dom shldr whl (armbld) ln 14"fat choic ckd rstcd
- grapes american type (slip skn) raw
- sauce barbecue
- girl scouts trefoils cookies
- pork ground 84 ln 16 fat ckd pan-broiled
- pizza hut 14" sausage pizza thin 'n crispy crust

=====

CLUSTER 7 SAMPLES

=====

- brussels sprouts ckd bld drnd wsalt
- pork ground 84 ln 16 fat ckd crumbles
- cookies graham crackers chocolate-coated
- beverages v8 splash smoothies peach mango
- gardenburger flame grilled burger frz unprep
- peas grn (includes baby lesuer types) cnd drnd sol unprep
- beverages lemonade frz conc pink prep w h2o
- frozen novelties no sugar added fudgsicle pops

- kraft velveeta past process chs sprd
- vela meen

=====

CLUSTER 8 SAMPLES

=====

- sorghum flr refined unenr
- danish pastry fruit unenr (incl appl cinn raisin strawberry)
- beef shrt loin t-bone steak lnfat 18"fat sel raw
- chicken breast deli rotisserie seasoned sliced prepackaged
- morningstar farms hotspicy veggie sausage patties frz unprep
- lamb nz imp frz leg whl (shksirl) ln fat 18" ckd rstd
- burdock root ckd bld drnd wsalt
- morningstar farms mushroom lover's burger frz unprep
- mother's iced lemonade cookies
- lamb dom shldr arm ln 14"fat choic ckd brld

=====

CLUSTER 9 SAMPLES

=====

- soybeans green raw
- campbell's red white tomato bisque cond
- beef loin tenderloin rst bnless ln fat 0" fat choic raw
- alcoholic bev wine table red cabernet sauvignon
- cookies pnut butter refr dough
- soy sau red na made from hydrolyzed veg prot
- turkey breast from whl bird meat only rstd
- cookies oatmeal sndwch w creme filling
- beef chuck eye steak bnless ln 0" fat sel raw
- oven-roasted chick breast roll

=====

CLUSTER 10 SAMPLES

=====

- lettuce red leaf raw
- campbell's chunky soups firehouse - hot spicy bf bean chili
- cherries sour red cnd lt syrup pk solliquids
- bev dairy drk mix choc red cal w low-cal sweeteners pdr
- asafoetida
- sweeteners tabletop fructose dry pdr
- pork ground 84 ln 16 fat raw
- yogurt fruit lofat 9 grams prot per 8 oz
- turkey back from whl bird meat only w added soln raw
- beef rnd tip rnd rst ln 0" fat sel ckd rstd

=====

CLUSTER 11 SAMPLES

=====

- banquet salisbury steak w gravy family size frz unprep
- bread french or vienna whl wheat
- pork sausage linkpatty fully ckd microwaved
- inf formula nestle good start 2 essentials w iron rtf
- keebler fudge shoppe fudge grahams 12 dipped red fat
- beef plate steak bnless inside skirt ln 0" fat sel ckd grild
- beef rib eye steak bone-in lip-on ln 18" fat choic ckd grld
- bread wheat germ toasted
- alcoholic bev whiskey sour prep from item 14028
- vegetable oil palm kernel

=====

CLUSTER 12 SAMPLES

=====

- soup bean w pork cnd prep w eq volume h2o
- crackers rusk toast
- broccoli frz spears unprep
- pork cured ham w nat juices slice bnless ln htd pan-broil
- pork frsh loin cntr rib (chops) bnless ln ckd brsd
- sausage smoked link sausage pork bf
- plum
- kiriyani
- lasagna w meat sau frz entree
- chokecherries raw pitted (northern plains indians)

=====

CLUSTER 13 SAMPLES

=====

- tortilla chips taco-flavor
- beans bkd cnd pln or vegetarian
- duck yng duckling dom wh pekin leg meatskn bone in ckd rstcd
- beef chuck clod rst ln only to 14" fat all grds ckd rstcd
- beef loin tenderloin rst ln bnless 0" fat sel ckd rstcd
- tuna lt cnd in oil wosalt drnd sol
- lamb aus frsh leg hindshank heel on bne-in ln ft 18" brsd
- potato chips cheese-flavor
- babyfood dinner vegham jr
- cheese mozzarella non-fat

=====

CLUSTER 14 SAMPLES

=====

- brinjal-4
- beef ground 75 ln meat 25 fat raw
- beef rnd eye of rnd rst bnless ln 0" fat all grds ckd rstcd
- kellogg's eggo mini muffin tops blueberry
- cheese tilsit
- jams and preserves

- pomfret snub nose
- yogurt fruit var non-fat
- fish salmon sockeye raw
- pork frsh loin cntr rib (chops or roasts) bone-in ln raw

=====

CLUSTER 15 SAMPLES

=====

- bologna bfpork lofat
- milk cnd cond swtnd
- cereals crm of rice dry
- cereals rte kellogg's frstd mini-wheats bite size strawb del
- turnips frz ckd bld drnd wsalt
- potatoes fr fr all types salt not added in proc frz ovn-htd
- beverages whey prot pdr isolate
- pork cured ham -- h2o added whl bnless ln unhtd
- squash smmr zucchini incl skn ckd bld drnd wsalt
- pie crust cookie-type prep fr recipe graham cracker chilled

=====

CLUSTER 16 SAMPLES

=====

- yogurt vanilla flavor lowfat milk swtnd w lo cal swtnr
- babyfood dssrt fruit pudd orange str
- crackers chs sandwich-type wpnut butter filling
- bacon bf stks
- morningstar farms buffalo wings frz unprep
- cowpeas catjang mature seeds ckd bld wsalt
- beverages tea hibiscus brewed
- cabbage red ckd bld drnd wosalt
- water chestnut
- sauce sofrito prep from recipe

=====

CLUSTER 17 SAMPLES

=====

- muffins blueberry toaster-type
 - walrus meat dried (alaska native)
 - cereals rte quaker cap'n crunch's oops! all berries crl
 - pizza hut 12" pepperoni pizza pan crust
 - chili con carne wbns cnd entree
 - beef ground 93 ln meat 7 fat crumbles ckd pan-browned
 - toaster pastries kellogg kellogg's pop tarts frstd raspberry
 - alcoholic bev liqueur coffee 53 proof
 - beef loin top srl cp stk bnls ln ft 18" ft algds ckd grld
 - pretzels hard pln made wunenr flr unsalted
- =====

CLUSTER 18 SAMPLES

=====

- restaurant latino black bean soup
- lamb var meatsby-products tongue raw
- beef lungs
- toaster pastries kellogg kellogg's pop tarts frstd cherry
- veal shldr blade chop ln fat ckd grilled
- cereals rte kellogg kellogg's spl k multi-grain
- french toast frz rth
- fish mahimahi raw
- beef top sirloin steak ln 18" fat sel ckd brld
- cheese brick

=====

CLUSTER 19 SAMPLES

=====

- pork frsh loin tp ln (chps) bnlss ln ft w added sln raw
- catsup low sodium
- ground turkey 93 ln 7 fat pan-broiled crumbles
- snacks mm mars kudos whl grain bars choc chip
- dessert topping semi solid frz
- keebler 100 cal right bites fudge shoppe mini brownies
- cookies choc sndwch wex creme filling
- sweet potato ckd bld wo skn
- cereals rte kashi golean crunch!
- lamb dom shldr arm lnfat 14"fat choic ckd brsd

=====

CLUSTER 20 SAMPLES

=====

- chickn broilr or fryrs brst sknlss bnless meat only ckd brsd
- egg poultry yolk raw
- turkey rtl parts breast meat only w added soln ckd rstcd
- sauce white thin prepared-from-recipe w butter
- chicken meatless
- pork sausage rice links brownserve ckd
- game meat muskrat raw
- candies reese's fast break milk choc pnt btr sft nugts
- lamb nz imp flap bnless ln ckd brsd
- cheese sub mozzarella

=====

CLUSTER 21 SAMPLES

=====

- salad drsng italian drsng red fat wo salt
- spinach frz chopd or leaf unprep
- epazote raw
- broccoli frz spears ckd bld drnd wosalt

- lamb nz imp frz comp of rtl cuts ln fat 18" fat raw
- burger king croissan'wich w sausage egg chs
- yellow rice w seasoning dry packet mix unprep
- sauce pnut made from cocnt h2o sugar pnuts
- salmon atlantic farmed raw
- campbell's chunky soups manhattan clam chowder

=====

CLUSTER 22 SAMPLES

=====

- beef ground 97 ln meat 3 fat patty ckd pan-broiled
- bread wheat sprouted
- soup clam chowder new england cnd rts
- sunshine krispy soup oyster crackers (large)
- shrimp mxd sp imitn made from surimi
- soup crm of onion cnd cond
- walrus liver raw (alaska native)
- keebler fudge shoppe jumbo fudge stks pnut butter
- banana ripe montham
- beef rib eye sml end (ribs 10- 12) ln 0" fat sel ckd brld

=====

CLUSTER 23 SAMPLES

=====

- syrups corn dk
- mcdonald's spicy buffalo sau
- lemonade frz conc white prep wh2o
- halibut greenland ckd dry heat
- potatoes hash brown frz w butter sau unprep
- cereals rte kellogg kellogg's spl k red berries
- cheese mexican queso cotija
- beef comp of rtl cuts ln fat 18" fat sel ckd
- chicken broilers or fryers meat only rstd
- murray sugar free fudge dipped mint cookies

=====

CLUSTER 24 SAMPLES

=====

- beef plate stk bnls otsde skrt ln fat 0" fat sel ckd grild
- pork frsh loin tenderloin ln ckd brld
- fish salmon king (chinook) raw (alaska native)
- beef chuck mock tender steak bnless ln 0" fat all grds raw
- waffles pln prep from recipe
- beverages arizona tea rtd lemon
- milk choc fluid comm red fat
- pistachio nuts dry rstd wosalt
- campbell's homestyle hr mexican style chick tortilla
- beef loin top loin ln fat 18" fat sel raw

=====

CLUSTER 25 SAMPLES

=====

- pepperidge farm farmhouse sourdough bread
- pork ground 72 ln 28 fat ckd pan-broiled
- babyfood veg carrots jr
- cream puff shell prep from recipe
- peppers hot chili red raw
- archway home style cookies coconut macaroon
- beverages coffee ready to drk milk bsd swtnd
- beef rib eye sml end (ribs 10-12) ln fat 0" sel ckd brld
- salmon pink ckd dry heat
- beef rnd eye of rnd rst ln 18" fat all grds ckd rstd

=====

CLUSTER 26 SAMPLES

=====

- chicken broilers or fryers dk meat thigh meat only raw
- cereals rte general mills fiber one bran crl
- broccoli frz spears ckd bld drnd wsalt
- fast foods biscuit wham
- milk filled fluid wlauric acid oil
- ostrich ground ckd pan-broiled
- olive garden spaghetti w meat sau
- beef rib eye rst bne-in lip-on ln 18" ft all grds ckd rstd
- lamb nz imp tunnel-boned lg chump off shank off ln fat raw
- mcdonald's pnuts (for sundaes)

=====

CLUSTER 27 SAMPLES

=====

- pork frsh loin cntr rib (chops) bnless lnfat ckd brsd
- campbell's 98 fat free broccoli chs soup cond
- chicken broilers or fryers breast meat skn ckd rotisserie
- pepperidge farm goldfish bkd snack crackers original
- fast foods sub swt onion chick wit brd w ltte tmt swt onion
- gravy brown dry
- sauce pizza cnd rts
- beef nz imp rump centre ln raw
- popcorn oil-popped lofat
- beef top loin filet bnless ln 18" fat choic ckd grilled

=====

CLUSTER 28 SAMPLES

=====

- cereals rte general mills cinn burst cheerios
- egg whl dried stabilized glucose red

- pork frsh loin cntr rib (chops or roasts) bnless ln raw
- kfc fried chick ex crispy wing meat skn w breadding
- pears raw
- froz novelties ice type sugar free ornge chry grp popsicle
- mori-nu tofu silken lite firm
- cake snk cakes not choc w icing or filling l-f w add fiber
- beef ground 90 ln meat 10 fat loaf ckd bkd
- beverages carb sprite lemon-lime wo caffeine

=====

CLUSTER 29 SAMPLES

=====

- rice white long-grain parbld unenr ckd
- jerusalem-artichokes raw
- goat chops
- chicken broilers or fryers thigh meat only ckd rstd
- broccoli frz chopd ckd bld drnd wosalt
- fish whitefish eggs (alaska native)
- pork cured ham h2o product rump bone-in ln fat htd rstd
- ostrich inside leg raw
- barley
- peppers hungarian raw

=====

CLUSTER 30 SAMPLES

=====

- campbell's chick w rice soup cond
- pork fresh ground cooked
- cream sub pdr lt
- crisped rice bar almond
- fish tuna lt cnd in h2o drnd sol
- beef comp of rtl cuts ln fat 0" fat all grds raw
- sunfish pumpkin sd ckd dry heat
- beef loin tenderloin rst bnless ln fat 0" fat sel ckd rstd
- beverages chocolate-flavored drk whey milk bsd
- pork frsh fat ckd

=====

CLUSTER 31 SAMPLES

=====

- beverages energy drk citrus
- english muffins whl grain white
- leavening agents crm of tartar
- red snapper
- burger king croissan'wich w sausage chs
- milkfish raw
- turkey dk meat from whl meat skn w added soln ckd rstd
- beef au imp wgu tdlm stkrst bnles ln fat mrb scr 45 raw

- cake boston crm pie commly prep
- gogu leaves green

=====

CLUSTER 32 SAMPLES

=====

- cowpea brown
- kashi chick florentine frz entree
- beverages energy drk red bull sugar free w added caffeine
- rye flour dark
- cereals cream of wheat inst dry
- babyfood meat meat stks jr
- candies gumdrops starch jelly pieces
- lima bns immat seeds frz fordhook unprep
- macaroni or noodles w chs made from red fat packaged mix un
- soup turkey chunky cnd rts

=====

CLUSTER 33 SAMPLES

=====

- loquats raw
- brinjal-17
- cauliflower leaves
- macaroni chs dinner w dry sau mix boxed unckd
- beef aus imp wgu top ln stkrst bnless ln mrb scr 45 raw
- gardenburger black bean chipotle burger frz unprep
- cake yel commly prep w choc frstng in-store bakery
- mangos raw
- babyfood dinner vegbacon str
- gravy franco-american bf slow rst fat free

=====

CLUSTER 34 SAMPLES

=====

- rolls hamburger or hotdog red-cal
- egg duck whole raw
- beverages cranberry-grape juc drk btld
- keebler swt cremes cookies
- beef nz imp flank ln fat ckd brsd
- fish pollock alaska raw (may have been previously frozen)
- campbell's homestyle mexican style chick tortilla soup
- sea lion steller meat w fat (alaska native)
- keebler toast pnut butter sndwch crackers
- corn wredgrn peppers cnd solliquids

=====

CLUSTER 35 SAMPLES

=====

- knol - khol
- domino's 14" sausage pizza crunchy thin crust
- beef rnd top rnd rst bnless ln 0" fat sel raw
- snacks general mills betty crocker frt rl ups bry flv wvt c
- pork frsh spareribs lnfat ckd brsd
- mace ground
- morningstar farms grillers qrtr pnd veggie brgr frz unprp
- murray honey graham
- campbell's 25 less na chick noodle soup cond
- salmon sockeye cnd wosalt drnd sol wbone

CLUSTER 36 SAMPLES

- jujube chinese frsh dried
- dates dry pale brown
- fast foods submarine sndwch cold cut brd w lettuce tomato
- allathi
- guavas common raw
- spot raw
- fruit leather rolls
- beef var meatsby-products lungs ckd brsd
- beef shldr pot rst or steak bnless ln 0" fat choic raw
- chicken broilers or fryers leg meat only ckd fried

CLUSTER 37 SAMPLES

Even after letting the algorithm separate data into 55 clusters, most of the clusters do not make sense. HDBSCAN is partially effective on this dataset, but it does not produce clean semantic food categories. Instead, it identifies density-based nutritional similarity regions, which often mix semantically different foods that share similar macro- or micro-nutrient profiles. This is especially visible in clusters combining snacks, grains, and processed foods.

Given the structure of the dataset (high-dimensional nutritional features with strong correlation between variables like fat, calories, and carbs), HDBSCAN is appropriate for discovering hidden structure and outliers, but *not* ideal for producing human-interpretable categorical groups.

Cluster	Main Theme	Representative Foods	Why It Works Well
8	Fast food & high-fat processed meals	Salisbury steak, sausage, whiskey sour, palm oil	Strong grouping of calorie-dense processed foods and fats.
12	Refined carbohydrates & processed foods	Pizza, crackers, potatoes, cereals	Clear separation around processed carb-heavy foods.
13	Lean meats & savory protein foods	Chicken breast, ham, crayfish, gravy	Consistent high-protein savory products.
16	Bakery & snack foods	Pie crust, tortilla chips, pretzels, coffee mixes	Well-defined snack/refined grain cluster.

Cluster	Main Theme	Representative Foods	Why It Works Well
17	Red meat & animal protein	Beef ribs, turkey, fried chicken	Strong animal-protein-focused cluster.
19	Protein-rich prepared foods	Chicken, sausage, egg yolk, mozzarella	Cohesive processed protein/meat category.
24	Sauces, syrups & prepared meals	Corn syrup, buffalo sauce, hash browns	Distinct condiment and processed-food profile.
27	Savory cooked foods & nuts	Turkey, pasta sauce, tortillas, almonds	Relatively coherent prepared-meal cluster.
31	Oils, sauces & cooked proteins	Margarine, beef ribeye, pasta sauce	Clear high-fat and prepared-food orientation.
54	Fast food & salty processed foods	Cheeseburgers, tartar sauce, plantain chips	Strong clustering of processed restaurant-style foods.

3.4 KMeans (Tomko)

I chose the K-Means algorithm, an unsupervised machine learning technique designed for exploratory data analysis and clustering. The goal of this model is to partition the available dataset into a predefined number of clusters (k) such that the samples within a single cluster are as similar to each other as possible, while being as distinct as possible from samples in other clusters. In our case, similarity is measured using Euclidean distance within an n -dimensional space of nutritional features.

During the learning (training) phase, the model first initializes k initial cluster centers (centroids) using the optimized `k-means++` method. It then iteratively alternates between two steps: first, it assigns each data point to the nearest centroid, and second, it recalculates the coordinates of each centroid as the arithmetic mean of all the data points currently assigned to it. This process of minimizing the within-cluster variance is repeated until the positions of the centroids stop changing (i.e., the model converges). The result of the training is a set of k centroids representing the typical profile of a given group (e.g., the mathematical “ideal” of a vegetable or a sweet), along with an assignment vector that gives each food item in the dataset its specific cluster label.

```
[20]: # Define the range of k values to test
k_values = range(4, 16)

silhouette_scores = []
inertias = []
models = {}

print("Evaluating KMeans for different values of k...")

# The elegant for-loop approach
for k in k_values:
    # Initialize and fit the model
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42,
    ↪n_init="auto")
    labels = kmeans.fit_predict(X_preprocessed)
```

```

# Store the metrics
silhouette_scores.append(silhouette_score(X_preprocessed, labels))
inertias.append(kmeans.inertia_)

# Save the model in case this k turns out to be the best
models[k] = kmeans

print(f"k={k:2d} | Silhouette: {silhouette_scores[-1]:.4f} | Inertia:↳
↳{inertias[-1]:.0f}")

# Find the best k based on the highest Silhouette Score
best_k = k_values[np.argmax(silhouette_scores)]
best_kmeans_model = models[best_k]

print(f"\nOptimal number of clusters (k) based on Silhouette Score: {best_k}")
fig, ax1 = plt.subplots(figsize=(10, 6))

# Plot Silhouette Score on the left Y-axis
color = 'tab:blue'
ax1.set_xlabel('Number of Clusters (k)')
ax1.set_ylabel('Silhouette Score', color=color)
ax1.plot(k_values, silhouette_scores, marker='o', color=color, linewidth=2)
ax1.tick_params(axis='y', labelcolor=color)
ax1.axvline(x=best_k, color='r', linestyle='--', label=f'Best k={best_k}')

# Create a secondary Y-axis for Inertia (Elbow Method)
ax2 = ax1.twinx()
color = 'tab:orange'
ax2.set_ylabel('Inertia (Squared Distances)', color=color)
ax2.plot(k_values, inertias, marker='s', linestyle=':', color=color, linewidth=2)
ax2.tick_params(axis='y', labelcolor=color)

# Formatting
fig.tight_layout()
plt.title('KMeans Evaluation: Silhouette Score & Inertia (Elbow Method)')
ax1.grid(True, alpha=0.3)
ax1.legend(loc='upper left')
plt.xticks(k_values)
plt.show()

```

Evaluating KMeans for different values of k...

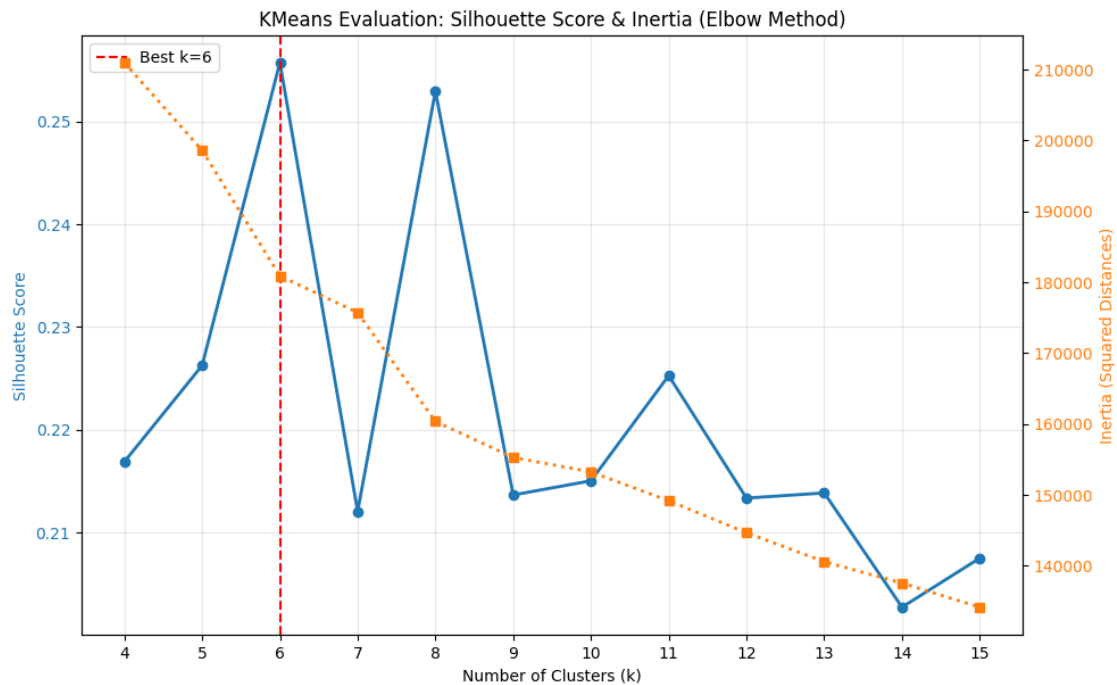
```

k= 4 | Silhouette: 0.2169 | Inertia: 210967
k= 5 | Silhouette: 0.2262 | Inertia: 198594
k= 6 | Silhouette: 0.2557 | Inertia: 180853
k= 7 | Silhouette: 0.2120 | Inertia: 175711
k= 8 | Silhouette: 0.2529 | Inertia: 160328
k= 9 | Silhouette: 0.2136 | Inertia: 155253

```

k=10 | Silhouette: 0.2150 | Inertia: 153216
 k=11 | Silhouette: 0.2253 | Inertia: 149143
 k=12 | Silhouette: 0.2133 | Inertia: 144623
 k=13 | Silhouette: 0.2138 | Inertia: 140544
 k=14 | Silhouette: 0.2027 | Inertia: 137514
 k=15 | Silhouette: 0.2075 | Inertia: 134123

Optimal number of clusters (k) based on Silhouette Score: 6



3.4.1 Best Model

We visualize the data via dimensionality reduction and compute medians for all features. Furthermore, we also pick a few ingredients from each cluster at random, to get a better picture of what the model learned.

```
[21]: # Reattach original dataframe
df_results = X_eng.copy()

df_results['Cluster'] = best_kmeans_model.labels_
df_results['Descrip'] = df.loc[df_results.index, 'Descrip']
df_results['NDB_No'] = df.loc[df_results.index, 'NDB_No']

# Reorder the columns
cols = ['NDB_No', 'Descrip', 'Cluster'] + [c for c in df_results.columns if c_
↳ not in ['NDB_No', 'Descrip', 'Cluster']]
```

```

df_results = df_results[cols]

# Print 10 random foods from each cluster:
for i in range(df_results['Cluster'].nunique()):
    print("\n=====")
    print(f"CLUSTER {i} SAMPLES")
    print("=====")

    # Min in case a cluster is tiny
    sample_size = min(10, len(df_results[df_results['Cluster'] == i]))
    samples = df_results[df_results['Cluster'] == i]['Descrip'].
    ↪sample(sample_size, random_state=42)

    for food in samples.values:
        print(f"- {food}")

```

```

=====
CLUSTER 0 SAMPLES
=====
- kellogg's eggo biscuit scramblers egg chs
- pepperidge farm goldfish bkd snack crackers cheddar
- pork fresh backfat raw
- cereals rte kashi heart to heart oat flakes bluebry clstrs
- rice brown parbld dry uncle ben's
- salad drsng mayo imitn soybn
- kellogg's eggo waffles blueberry
- tgi friday's chick fingers from kids' menu
- sauce pesto buitoni pesto w basil rts refr
- frozen novl klondike slim-a-bear no sgr added stickless bar

```

```

=====
CLUSTER 1 SAMPLES
=====
- silan
- parcus
- sankata paarai
- gold fish
- kalava
- crab sea
- vora
- oyster
- allathi
- kite fish

```

```

=====
CLUSTER 2 SAMPLES
=====

```

- frankfurter bf lo fat
- game meat elk raw
- chicken broilers or fryers thigh meatskn ckd fried flr
- beef aus imp grss-fed loin tenderln stkrst bnless raw
- usda commodity bf patties wvpp frz raw
- beef flank steak ln 0" fat all grds raw
- chicken broilers or fryers back meat only ckd stwd
- beef ground 93 ln meat 7 fat
- trout mixed species raw
- croaker atlantic ckd breadedfried

=====

CLUSTER 3 SAMPLES

=====

- spices thyme dried
- cereals rte kellogg's rice krispies gluten free
- cereals rte general mills berry berry kix
- beverag unilev slimfast shake hi prot whey pdr 3-2-1 plan
- cereals rte quaker quaker oat life pln
- babyfood crl oatmeal w bananas dry
- cereals rte malt-o-meal blueberry muffin tops crl
- cereals rte general mills kix
- babyfood crl oatmeal whoney dry
- pork frsh var meatsby-products liver ckd brsd

=====

CLUSTER 4 SAMPLES

=====

- cereals rte kellogg kellogg's raisin bran
- beverages tea inst unswtnd pdr
- leavening agents baking pdr double-acting straight po4
- cereals quaker inst oatmeal banana bread dry
- rye grain
- cereals crm of wheat 2 12 minute cook time dry
- amaranth grain unckd
- beverages tea inst swtnd w na saccharin lemon-flavored pdr
- safflower sd krnls dried
- almonds dry rstdd wosalt

=====

CLUSTER 5 SAMPLES

=====

- grapes cnd thompson seedless h2o pk solliquids
- babyfood potatoes toddler
- broadbeans (fava bns) mature seeds cnd
- sauce barbecue kc masterpiece original
- caribou liver raw (alaska native)
- babyfood fruit applsauc jr

- morningstar farms brkfst pattie w organic soy frz unprep
- campbell's crm of mushroom soup cond
- butterbur (fuki) raw
- clam mxl sp cnd liq

```
[22]: # Name clusters, plot again

cluster_names = {
    0: "Processed Snacks & Sweets",      # High cal, carbs, fat, low fiber
    1: "Seafood & Organ Meats",         # Low cal, pure protein, extreme vitamin
    2: "Standard Meats & Poultry",       # High protein, moderate fat, zero carb
    3: "Fortified Cereals & Carbs",      # High carb, sugar, extreme mineral/
    4: "Grains, Nuts & Seeds",           # Moderate protein, high complex carbs,
    5: "Fruits, Veggies & Beverages"    # Low cal, low fat/protein, high water
}

# Map the numeric clusters to their new names
df_results['Cluster_Name'] = df_results['Cluster'].map(cluster_names)

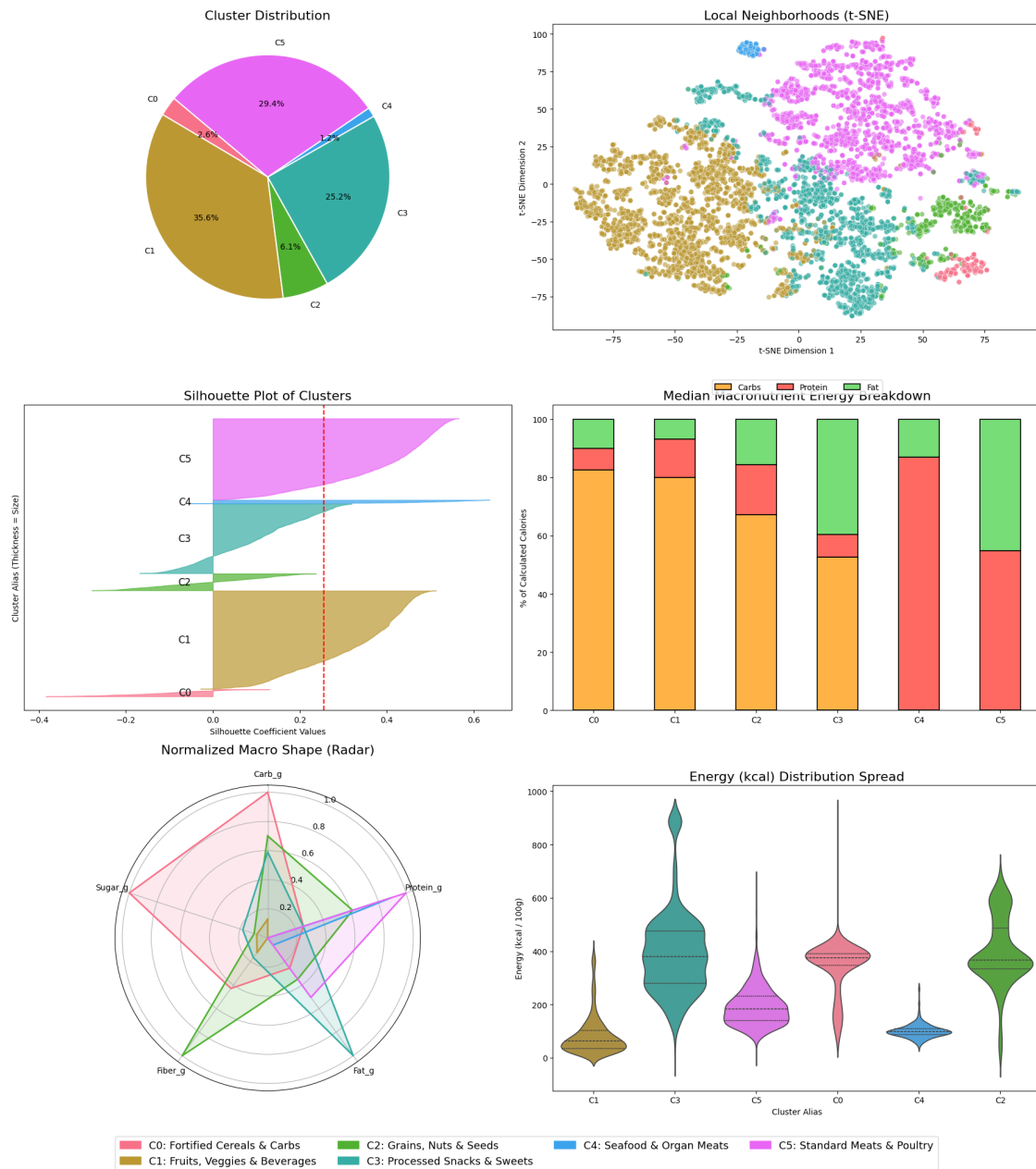
# Reorder columns to make it highly readable
cols = ['NDB_No', 'Descrip', 'Cluster', 'Cluster_Name'] + \
    [c for c in df_results.columns if c not in ['NDB_No', 'Descrip',
    'Cluster', 'Cluster_Name']]
df_results = df_results[cols]

evaluate_and_visualize_clusters(X_eng, X_preprocessed,
    df_results['Cluster_Name'])
```

CLUSTER SIZE STATISTICS

Cluster	Alias	Size (Rows)	Percentage (%)
Fortified Cereals & Carbs	C0	237	2.59
Fruits, Veggies & Beverages	C1	3256	35.56
Grains, Nuts & Seeds	C2	556	6.07
Processed Snacks & Sweets	C3	2304	25.16
Seafood & Organ Meats	C4	114	1.25
Standard Meats & Poultry	C5	2689	29.37

Comprehensive Cluster Analysis Dashboard



3.4.2 Subclustering

The clustering was a success, however, the largest cluster still makes up more than a third of the original data. Thus we've decided to subcluster the largest cluster (again, with KMeans). - K=5 produced good results, although one cluster inevitably collapses.

```
[23]: # Isolate the scaled data and the engineered data for just Cluster 5
cluster_5_indices = df_results[df_results['Cluster'] == 5].index

X_preprocessed_c5 = X_preprocessed.loc[cluster_5_indices]
df_c5_results = df_results.loc[cluster_5_indices].copy()

# Hypothesis: it will split into Vegetables, Sugary Fruits/Juices, and Zero-Cal
↳Beverages, ...
k_sub = 5
kmeans_sub = KMeans(n_clusters=k_sub, init='k-means++', random_state=42,
↳n_init="auto")
df_c5_results['Sub_Cluster'] = kmeans_sub.fit_predict(X_preprocessed_c5)

# Rename clusters for better visualizations
sub_cluster_names = {
    0: "Starchy Veggies & Soups",
    1: "Fibrous & Green Veggies",
    2: "Vitamin Outliers (Acerola)",
    3: "Sugary Drinks, Juices & Alcohol",
    4: "Dairy Liquids, Formulas & Creams"
}

df_c5_results['Sub_Cluster_Name'] = df_c5_results['Sub_Cluster'].
↳map(sub_cluster_names)

evaluate_and_visualize_clusters(X_eng.loc[cluster_5_indices], X_preprocessed.
↳loc[cluster_5_indices], df_c5_results['Sub_Cluster_Name'])

# Print random samples from each sub-cluster
print("\n" + "="*50)
print("SUB-CLUSTER SAMPLES")
print("="*50)
for i in range(k_sub):
    print(f"\n--- Sub-Cluster {sub_cluster_names[i]} ---")
    sub_cluster_data = df_c5_results[df_c5_results['Sub_Cluster'] == i]

    # Get the sample size based on the actual size of this specific sub-cluster
    sample_size = min(8, len(sub_cluster_data))
    samples = sub_cluster_data['Descrip'].sample(sample_size, random_state=42)

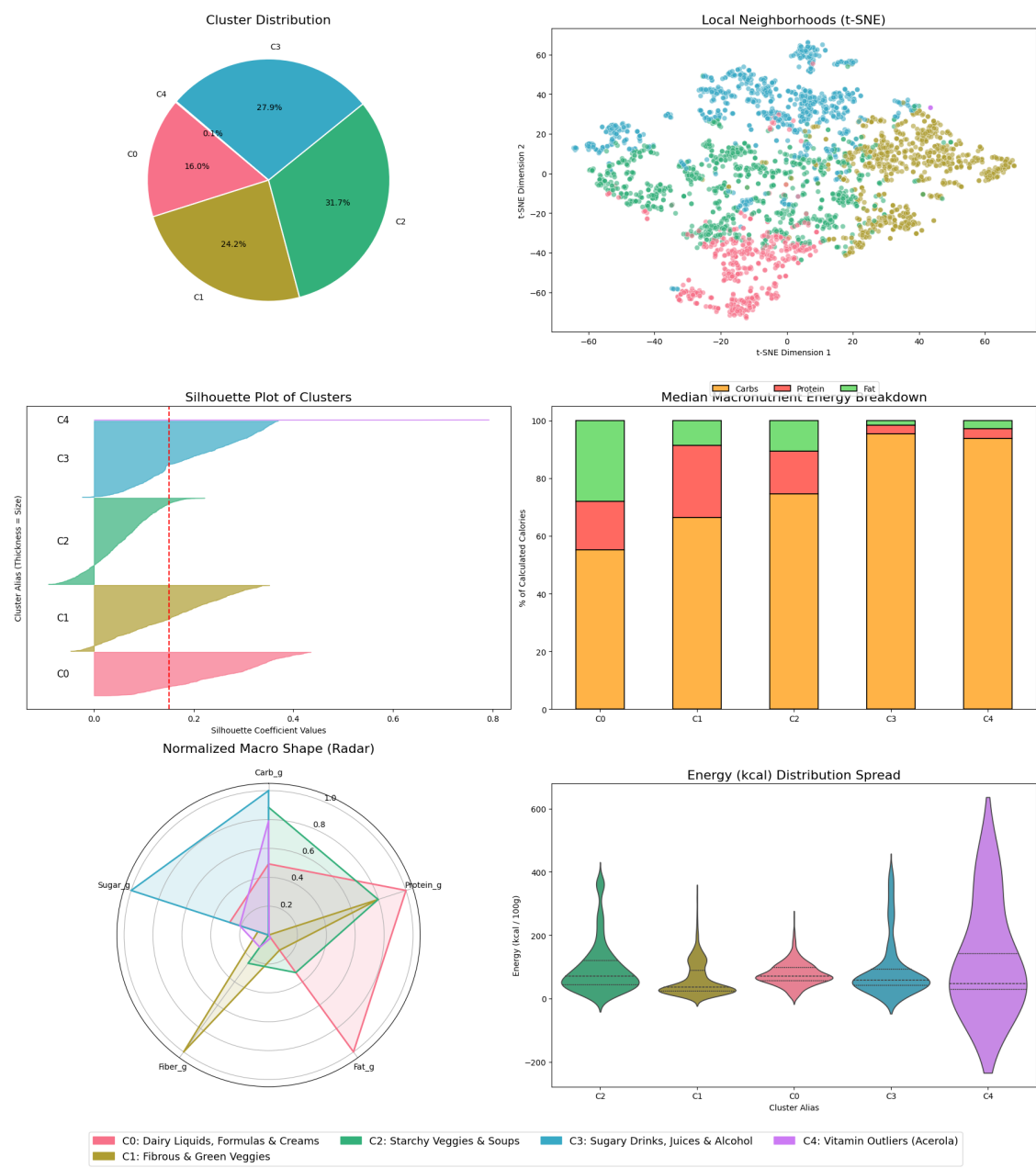
    for food in samples.values:
        print(f"    - {food}")
```

```
=====
CLUSTER SIZE STATISTICS
=====
```

```
Alias    Size (Rows)    Percentage (%)
```

Cluster			
Dairy Liquids, Formulas & Creams	C0	521	16.00
Fibrous & Green Veggies	C1	789	24.23
Starchy Veggies & Soups	C2	1032	31.70
Sugary Drinks, Juices & Alcohol	C3	910	27.95
Vitamin Outliers (Acerola)	C4	4	0.12

Comprehensive Cluster Analysis Dashboard



=====

SUB-CLUSTER SAMPLES

=====

- Sub-Cluster Starchy Veggies & Soups ---
 - soup clam chowder manhattan cnd cond
 - peppers hot chili red cnd excluding seeds solliquids
 - seaweed irishmoss raw
 - corn swt white cnd vacuum pk reg pk
 - winged bean immat seeds ckd bld drnd wsalt
 - wheat flr white (industrial) 15 prot bleached unenr
 - horned melon (kiwano)
 - tky stuffing mshd potato wgravy assorted veg frz microwave
- Sub-Cluster Fibrous & Green Veggies ---
 - celery stalk
 - cabbage savoy raw
 - broccoli frz chopd unprep
 - babyfood peas dices todd
 - edamame frz unprep
 - okra ckd bld drnd wsalt
 - cardoon ckd bld drnd wsalt
 - peppers serrano raw
- Sub-Cluster Vitamin Outliers (Acerola) ---
 - acerola (west indian cherry) raw
 - babyfood gerber 2nd foods appl carrot squash organic
 - acerola juice raw
 - beverages tea grn inst decaffei lemon unswtnd fort w vit c
- Sub-Cluster Sugary Drinks, Juices & Alcohol ---
 - guava nectar cnd w added vit c
 - beverages carb tonic h2o
 - candies jellybeans
 - alcoholic bev wine table red zinfandel
 - alcoholic bev whiskey sour prep from item 14028
 - soymilk choc w added ca vitamins a d
 - beverages fruit juc drk red sugar w vitamin e added
 - grapefruit juc white cnd or btld unswtnd
- Sub-Cluster Dairy Liquids, Formulas & Creams ---
 - inf formula nestle good start soy w dha ara rtf
 - inf formula abbo nu simil go gr rtf w ara dha
 - prego pasta garlic supreme ital sau rts
 - light ice crm creamsicle

- beverages coffee sub crl grain bev pdr prep w whl milk
- babyfood ravioli chs filled wtomato sau
- kefir lowfat strawberry lifeway
- rennin vanilla dry mix prep w whl milk

3.4.3 Interpretation

Based on the hidden mathematical patterns in the data, our model learned to recognize natural human dietary categories with surprising accuracy. An analysis of the learned parameters (the centroid coordinates) reveals that the model did not group foods randomly; instead, it created clusters that mirror real macrobiotic and physical properties. For example, it successfully separated foods with high water content and low energy density (fruits, vegetables) from energy-dense and highly processed foods.

The reason why the model learned exactly this is a direct result of our feature engineering and pre-processing pipeline. The model was highly sensitive to our derived columns, such as the macronutrient calorie ratios and the engineered Fiber_Carb_ratio and Vitamin_density features. Standard K-Means is notoriously sensitive to extreme outliers (e.g., a food with a massive amount of sodium could hijack a centroid and create a meaningless cluster). Because we applied a logarithmic transformation (log1p) to the highly right-skewed mineral distributions and normalized everything using StandardScaler, the model treated all key nutrients with equal mathematical weight. This allowed it to subtly differentiate between “Fibrous & Green Veggies” and “Starchy Veggies,” or to identify the tiny “Seafood & Organ Meats” cluster defined by zero carbohydrates but extreme vitamin density.

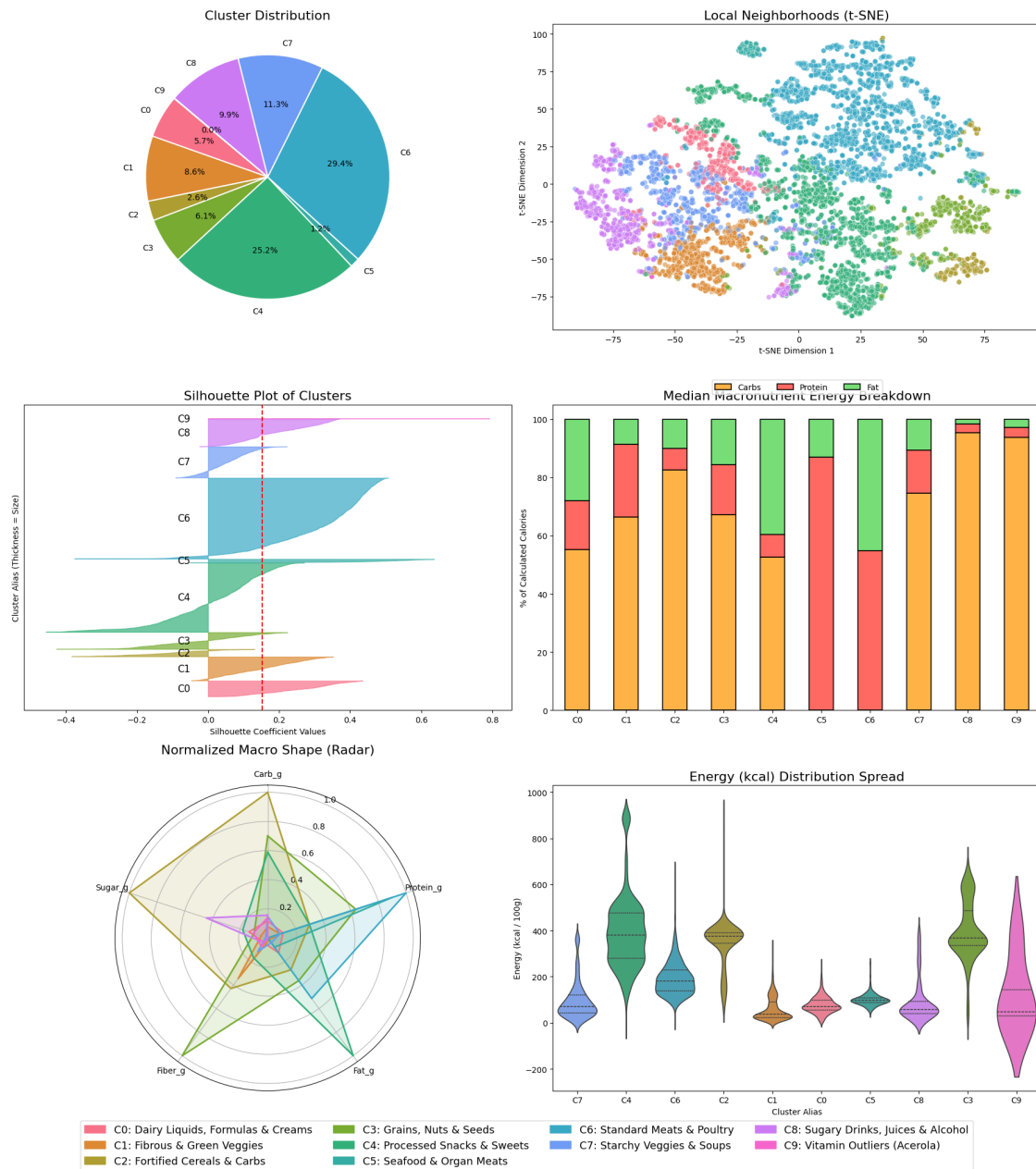
```
[24]: # For rows that aren't in Cluster 5, keep their original Cluster_Name.
# For rows that ARE in Cluster 5, overwrite it with the new Sub_Cluster_Name.
df_results.update(df_c5_results[['Sub_Cluster_Name']].
    ↳rename(columns={'Sub_Cluster_Name': 'Cluster_Name'}))

evaluate_and_visualize_clusters(X_eng, X_preprocessed,
    ↳df_results['Cluster_Name'])
```

```
=====
```

CLUSTER SIZE STATISTICS			
=====			
Cluster	Alias	Size (Rows)	Percentage (%)
Dairy Liquids, Formulas & Creams	C0	521	5.69
Fibrous & Green Veggies	C1	789	8.62
Fortified Cereals & Carbs	C2	237	2.59
Grains, Nuts & Seeds	C3	556	6.07
Processed Snacks & Sweets	C4	2304	25.16
Seafood & Organ Meats	C5	114	1.25
Standard Meats & Poultry	C6	2689	29.37
Starchy Veggies & Soups	C7	1032	11.27
Sugary Drinks, Juices & Alcohol	C8	910	9.94
Vitamin Outliers (Acerola)	C9	4	0.04

Comprehensive Cluster Analysis Dashboard



3.4.4 Evaluation

To evaluate the clustering quality, we utilized the Silhouette Score complemented by an analysis of Inertia (Within-Cluster Sum of Squares). Inertia measures the compactness of the clusters; however, it naturally decreases as the number of clusters increases, making it insufficient on its own. The Silhouette Score is much more appropriate for this task because it measures (on a scale from -1 to 1) how close each data point is to its own cluster compared to the nearest neighboring cluster. It

therefore accounts for both cluster cohesion and cluster separation. Our choice of k was guided by finding the maximum of this score in combination with the “elbow” method applied to the inertia graph.

```
[25]: # Silhouette Score
# Measures how similar an object is to its own cluster compared to other
# → clusters.
# Ranges from -1 to +1. A higher score indicates well-separated, dense clusters.
trained_silhouette = silhouette_score(X_preprocessed, df_results['Cluster_Name'])

print(f"Final Number of Clusters (k) : {len(df_results['Cluster_Name']).
# → unique()})")
print(f"Silhouette Score : {trained_silhouette:.4f}")
```

```
Final Number of Clusters (k) : 10
Silhouette Score : 0.1515
```

4 Conclusion

4.1 Summary

This project successfully developed a nutritional clustering system to organize food ingredients into semantically meaningful categories based on their macronutrient and micronutrient profiles. The comprehensive pipeline included rigorous data preprocessing, exploration of multiple clustering algorithms, and careful hyperparameter optimization.

4.1.1 Methodology

Data Preprocessing: We began with 10250 manually cleaned food records from the USDA nutritional database. After handling missing values, converting micro/milligrams to grams, removing outliers that violated the Atwater caloric equation and removing duplicate rows, we retained 9156 samples with 27 numeric features. Log1p transformation was applied to highly skewed distributions, followed by standard scaling to ensure equal feature weighting. (both are standard techniques in nutritional science)

Feature Engineering: We created derived features including macronutrient-calorie ratios, fiber-to-carb ratios, saturated fat ratios, and aggregated mineral/vitamin density scores. These engineered features enabled the model to capture domain-specific nutritional relationships beyond raw nutrient counts.

4.1.2 Algorithm Comparison:

- **Random Baseline:** Silhouette score = -0.0187 (trivial)
- **DBSCAN:** Good silhouette score ≈ 0.47 but producing only 4 poorly-interpreted clusters
- **HDBSCAN:** Great silhouette score ≈ 0.34 , 37 density-based clusters with mixed semantic coherence; effective for outlier detection but over-fragmented and hard to interpret.
- **K-Means:** Optimal $k=6$ with silhouette score ≈ 0.25 , producing highly interpretable clusters aligned with real dietary categories. Used further sub-clustering to cluster large clusters further and achieved better interpretable clusters with lower (≈ 0.16) silhouette score.